# Why Single-Site Samplers Underestimate the Number of Communities

Project P3.C in detail: Bayesian stochastic block models with an unknown number of communities,
the cost of opening a community, and what can be proved


**Section 1** Annotated literature relevant to P3.C

**Section 2** The project in detail: model, heuristics, evidence, targets, proof plan


Literature checked up to 22 October 2026. Companion to the P3 document.

**Badges.** [Q0] invariance only [Q1] plus irreducibility argued [Q2] ergodicity proved [Q3] quantitative bound **[F]** failure [tool] general theory [context] background [core] central to P3.C.

**Verification.** FT full text read Abs abstract or summary Sec secondary source Meta bibliographic metadata only † a detail is unverified.

# Contents

**Overview** **2**

**1 Literature relevant to P3.C** **4**
1.1 Scope, method and legend . . . 4
1.2 Summary: what the literature says about P3.C . . . 4
1.3 Bayesian SBMs with an unknown number of communities: models, priors and samplers . . . 5
1.4 Description length, physics-style inference and resolution limits . . . 8
1.5 Statistical theory: Bayesian posteriors and the number of communities . . . 9
1.6 Frequentist estimation and testing of the number of communities . . . 11
1.7 Thresholds with many or growing numbers of communities . . . 12
1.8 Mixing, hardness and metastability: rigorous templates . . . 14
1.9 Samplers that change $K$, and split–merge designs . . . 16
1.10 General Markov chain tools . . . 16
1.11 What could not be verified . . . 17

**2 Project P3.C in detail** **19**
2.1 Summary . . . 19
2.2 Model, posterior and samplers . . . 19
2.3 Heuristic analysis . . . 20
2.4 Numerical evidence . . . 22
2.5 Precise targets . . . 23
2.6 Proof strategy . . . 25
2.7 Relation to existing work . . . 26
2.8 Risks, fallbacks and the second direction . . . 27
2.9 Work plan and paper outline . . . 27
2.10 Extensions and open questions . . . 28

**A Derivations** **29**
A.1 The price of one new edge probability . . . 29
A.2 The opening log-odds . . . 30
A.3 The average and the maximum over nodes . . . 30
A.4 The nucleus . . . 31
A.5 From free energies to times . . . 32
A.6 The underestimation law . . . 33
A.7 Extra blocks at the truth . . . 34
A.8 Visible splits . . . 34
A.9 Spectral splits . . . 34

**B Simulation details** **36**
B.1 Model, priors and implementation . . . 36
B.2 Samplers . . . 36
B.3 Diagnostics . . . 36
B.4 Success criteria . . . 37
B.5 Experiments . . . 37
B.6 Hitting times . . . 37

B.7 Run counts and reproducibility . . . . . 38
B.8 Limitations . . . . . 39

# Overview

**The question.** Bayesian stochastic block models (SBMs) with a prior on the number of communities are fitted by MCMC samplers that move one node at a time. Practitioners report that these samplers underestimate the number of communities when it is large (D9), and that single-node moves "systematically fail" (D7). Others report the opposite failure, a heavily overestimated number retained by the sampler (B10). No theorem explains any of this. P3.C proposes one.

**The mechanism.** Suppose the sampler holds two communities $H$ and $R$ in one block $M$. To separate them, it must first move a single node of $H$ into a new block.

- *Cost.* The new block needs a new row of the connectivity matrix. With $K$ blocks present, these are $K$ new edge probabilities, each estimated from about $n/K$ trials. Under Beta priors each costs about $\frac{1}{2}\log(n/K)$ nats.
- *Gain.* None, if $H$ and $R$ have the same size and internal density and connect to the rest of the network in the same way, as in the planted partition model, or for sub-communities in a hierarchy. Then the single node's edges look exactly like the block's.
- *Consequence.* Opening costs a factor of about $n^{-K/2} \cdot n^{-1} = n^{-(K+2)/2}$, where the $1/n$ comes from a Dirichlet-process partition prior (an MFM prior with parameter $\gamma$ gives $n^{-1-\gamma}$). From a merged pair $K = K_0 - 1$. The expected number of single-site updates before $H$ and $R$ are separated should therefore grow asymptotically like $n^{(K_0+1)/2}$, a polynomial whose degree grows with the number of communities, while split–merge moves separate them at once.

**What the new simulations show** (Section 2.4).

- At fixed block size the cost of opening grows by 2.46 nats per extra community, against 2.43 predicted. At $K_0 = 4$ it grows by 2.55 per unit of $\log n$, against 2.45 predicted.
- The rate at which single-site Gibbs opens blocks scales as $n^{-1.52}$ and $n^{-1.94}$ for $K_0 = 2$ and 3, against $n^{-1.50}$ and $n^{-1.98}$ predicted. Hitting times grow more slowly at these sizes, like $n^{1.39}$ and $n^{1.56}$ updates. This is consistent with a newly opened block surviving more often in larger graphs, a pre-asymptotic effect whose cause is not yet isolated.
- Single-site Gibbs never separated an invisible merged pair for $K_0 \geq 4$, in 34 distinct runs with budgets of 30 to 100 sweeps.
- From one block it found only 1–3 communities, even after $8 \times 10^5$ updates with $K_0 = 16$.
- Spectral split–merge separated merged pairs within 1–16 sweeps. From one block it recovered all communities for $K_0 \leq 8$, and 14–15 of 16 for $K_0 = 16$, within 30 sweeps.
- When the two merged communities connect differently to the rest of the network, single-site Gibbs separated them at once. The obstruction is specific to *splits that single nodes cannot see*.

**What the literature provides** (Section 1).

- No convergence theory for any SBM sampler with unknown $K$.
- One local mixing theorem with $K$ known (M1).
- Statistical results for $K$: mostly frequentist consistency; on the Bayesian side, concentration for $K_0 \leq 3$ (B14), consistent posterior-odds tests (S4), a consistent penalised evidence (S5), and adaptive contraction without consistency for $K$ (S2).
- Thresholds for many communities (T15, T12).
- Metastability templates from Potts models and planted problems (M5, M7, M3).

In P3.C's regime polynomial-time spectral methods recover the communities. So the slowness is an artefact of the sampler, not a computational barrier of the problem.

**What P3.C targets** (Section 2.5).

- The exponent law, with matching bounds.
- Superpolynomial barriers when $K_0$ grows.
- An underestimation law from the default start.
- A dichotomy between visible and invisible splits.
- Polynomial split–merge.
- Posterior concentration with growing $K_0$, including a conjecture that Dirichlet-process SBMs are consistent for $K$.
- Local rapid mixing with random $K$.
- A general opening-cost principle.

# 1 Literature relevant to P3.C

## 1.1 Scope, method and legend

**Scope.** P3.C asks when single-site MCMC samplers on Bayesian SBM posteriors with an unknown number of communities fail to create communities, and what can be proved about it. The relevant literature has eight parts:

(i) Bayesian SBMs and relatives with unknown $K$, under any inference method (Section 1.3);

(ii) description-length and physics-style inference, including resolution limits (Section 1.4);

(iii) statistical theory of the Bayesian posterior and of the number of communities (Section 1.5);

(iv) frequentist estimation and testing of $K$ (Section 1.6);

(v) thresholds with many or growing numbers of communities (Section 1.7);

(vi) rigorous mixing, hardness and metastability results that can serve as proof templates (Section 1.8);

(vii) samplers that change $K$, including split–merge designs (Section 1.9);

(viii) general Markov chain tools (Section 1.10).

Entries already described in the P3 document are repeated briefly, so that this document is self-contained.

**Badges and verification.** The badges are those of the P3 document. A new tag, **[core]**, marks entries that P3.C builds on directly. FT means the full text was read, Abs the abstract or a summary, Sec a secondary source, and Meta bibliographic metadata only. A dagger † marks a detail that could not be checked.

**Search.** Searches and reading took place on 15–22 October 2026. They comprised four topic surveys for the P3 document, two new surveys for this document (Bayesian SBMs with unknown $K$; mixing, hardness and thresholds), and spot checks. Project Euclid, Wiley, Springer, Taylor & Francis, ScienceDirect, ACM DL, HAL and PMC pages were often blocked. For those papers the arXiv version, Crossref metadata or search listings were used. "Not found" means not found in these searches.

## 1.2 Summary: what the literature says about P3.C

1. **Every SBM sampler that changes $K$ has at most an informal convergence argument.** This covers the samplers of McDaid et al., Peixoto, Newman & Reinert, Riolo et al., Geng et al., Ludkin, Sanna Passino & Heard, Legramanti et al., Amongero & De Blasi, Lu et al. and Durante et al. All are at level **[Q0]** or **[Q1]**. Fully collapsed samplers on partitions are uniformly ergodic for free, because the state space is finite, so the open questions are entirely quantitative.

2. **Both directions of failure are documented, and neither is explained.**
   - *Too few communities.* Riolo et al. (D9): accurate up to about 20 groups and underestimates beyond, from random starts, while runs started at the planted partition recover $K$.
   - *Failures of single-node moves with no direction stated.*
     - Peixoto (D7): single-node moves "systematically fail".
     - Newman & Reinert (D8): metastable states.
     - Lee & Battiston (B22): "stuck at $K_n = 4$".
   - *Too many communities.* Côme & Latouche (B10): the collapsed sampler of McDaid et al. "highly overestimates" $K$, because moves that reduce complexity are rarely accepted.

   P3.C targets the first direction, where the barrier is a cost that grows with $K$. The second direction is discussed in Section 2.8.

3. **One quantitative theorem exists for a Bayesian SBM posterior**, with $K$ known: Zhuo & Gao's local rapid mixing from a warm start (M1). Nothing quantitative exists with $K$ unknown, upper or lower.

4. **Statistically, $K$ is well understood for frequentist procedures but much less so for posteriors.**
   - *Frequentist.* There are many consistent estimators of $K$ (Section 1.6).
   - *Bayesian.*
     - Geng et al. prove posterior consistency for $K$ only when $K_0 \in \{2, 3\}$ (B14).

  - Jiang & Tokdar need an assortative prior (B17).
  - Gao, van der Vaart & Zhou give adaptive contraction, not consistency for $K$ (S2).
  - A penalised Krichevsky–Trofimov evidence is consistent (S5).
  - Posterior-odds tests are consistent (S4).
- *Not found:* a consistency result for $K$ under a Dirichlet-process SBM.
- *Resolution limits.* Under flat description-length priors at most about $\sqrt{N}$ groups are detectable (D3). Nested priors push this towards $N/\log N$ (D4).

5. **Thresholds.** With many communities there are gaps between statistical and computational thresholds below the Kesten–Stigum bound (T12, T5, T15). P3.C works well above exact recovery, where spectral methods succeed in polynomial time (T3, T4, T9). *So the obstruction studied in P3.C is an artefact of single-site samplers, not a hardness of the problem.* This is a key point for the paper's message.

6. **Proof templates exist.**
   - Metastability at first-order transitions of Potts models: mean field (M5) and random regular graphs, with a planted-model conductance argument (M7).
   - Planted-clique bottlenecks for Metropolis (M3, M4).
   - Overlap gaps for modularity (M11).
   - Canonical paths for SBM posteriors (M1).
   - The local-majority characterisation of exact recovery (T7), which makes single-site moves locally stable at the truth.

| Ingredient of P3.C | Available from | Missing |
|---|---|---|
| Model and samplers with random $K$ | B14, B18, B9, D9, D6, X2 | any convergence theory |
| Evidence that $K$-changing moves fail | D9, D7, D8, B10, B22 | a mechanism and a theorem |
| Posterior concentration on $(K_0, C_\star)$ | B14 ($K_0 \leq 3$), B17 (assortative prior), S4, S5; S2 (contraction only) | standard Beta priors with growing $K_0$; DP priors |
| Local stability at the truth | T7, M1, T4 | random-$K$ version |
| Bottleneck and metastability arguments | M5, M7, M3, M4, M11, G1, G2 | the SBM opening cost and nucleus analysis |
| Polynomial upper bounds | M1 (local), G3, G4, X5 | global bounds with split–merge |
| Spectral splitting of blocks | F1, F9, G5 | analysis inside an MH move |

**Table 1:** Ingredients for P3.C and where they come from.

## 1.3 Bayesian SBMs with an unknown number of communities: models, priors and samplers

Each entry records how $K$ is treated, the inference method, any convergence statement, and any theory.

**B1** **Estimation and prediction for stochastic blockmodels for graphs with latent block structure** [$K$ known] [Q0] Abs

T. A. B. Snijders, K. Nowicki (1997). *J. Classification 14, 75–100. doi: 10.1007/s003579900004*

The origin of the Bayesian SBM. Two blocks, fixed. Small graphs are fitted by EM and large ones by Gibbs sampling. The blocks are recovered with probability tending to one as $n \to \infty$ with fixed edge probabilities. There is no convergence statement for the sampler.

**B2** **Estimation and prediction for stochastic blockstructures** [$K$ known] [Q0] Abs Sec

K. Nowicki, T. A. B. Snijders (2001). *J. Amer. Statist. Assoc. 96(455), 1077–1087*

Any number of blocks, fixed by the user, with different $K$ compared through summaries. Gibbs sampling, with label switching handled by invariant summaries. The manual gives "no definite guidelines" for burn-in.

**B3** **Learning systems of concepts with an infinite relational model** [$K$ random] [Q0] FT

C. Kemp, J. B. Tenenbaum, T. L. Griffiths, T. Yamada, N. Ueda (2006). *Proc. AAAI-06, 381–388*

The infinite relational model: CRP($\gamma$) partitions with Beta link probabilities integrated out. This is exactly the collapsed DP–Beta SBM posterior of P3.C. The results come from hill-climbing to the MAP partition with move, split and merge proposals. MCMC "can" be used (citing Jain–Neal).

**B4 Infinite hidden relational models** [$K$ random] [Q0] FT

Z. Xu, V. Tresp, K. Yu, H.-P. Kriegel (2006). *Proc. UAI 2006; arXiv:1206.6864*

A DP/CRP prior for each entity class. Uncollapsed Gibbs reassigns one entity at a time and opens a new cluster with probability proportional to $\alpha_0$, with fresh parameters drawn from $G_0$. The reported results come from a deterministic approximation. There is no convergence statement.

**B5 A Bayesian approach to network modularity** [$K$ unknown] [context] FT

J. M. Hofman, C. H. Wiggins (2008). *Phys. Rev. Lett. 100, 258701†; arXiv:0709.3512*

Variational Bayes for a two-parameter (within/between) SBM with Beta and Dirichlet priors. The number of occupied modules is learned: the occupation of surplus modules "converges to zero". The optimisation is local, so restarts are needed. There is no MCMC.

**B6 A mixture model for random graphs** [$K$ unknown] [context] Sec

J.-J. Daudin, F. Picard, S. Robin (2008). *Statist. Comput. 18†. doi:10.1007/s11222-007-9046-7*

Variational EM for a fixed number of blocks $Q$, with $Q$ chosen by an integrated classification likelihood (ICL) criterion. This is the standard description; the content and pages were not verified.

**B7 Variational Bayesian inference and complexity control for stochastic block models** [$K$ unknown] [context] Abs FT

P. Latouche, E. Birmelé, C. Ambroise (2012). *Statistical Modelling 12(1)†, 93–115; arXiv:0912.2873*

Variational Bayes with Jeffreys priors. $Q$ is chosen by maximising a non-asymptotic variational bound on the marginal likelihood (ILvb). In simulations with $N = 50$ nodes and six classes, ICL picks the right $Q$ 12% of the time and ILvb 73%; ICL underestimates $Q$.

**B8 Bayesian community detection; Nonparametric Bayesian modeling of complex networks: an introduction** [$K$ random] [Q0] Meta/FT

M. Mørup, M. N. Schmidt (2012); M. N. Schmidt, M. Mørup (2013) (2012, 2013). *Neural Comput. 24(9), 2434–2456; IEEE Signal Process. Mag. 30(3), 110–128 (doi:10.1109/MSP.2012.2235191; arXiv:1312.5889)*

For the first paper only the citation was confirmed. The second is a tutorial on the infinite relational model, with Beta–Bernoulli parameters integrated out. Its experiments run single-site collapsed Gibbs together with Jain–Neal split–merge, use five random restarts "to improve mixing", and run 250 burn-in plus 250 sampling iterations without diagnostics.

**B9 Improved Bayesian inference for the stochastic block model with application to large networks** [$K$ random] [Q1] FT [core]

A. McDaid, T. B. Murphy, N. Friel, N. J. Hurley (2013). *Comput. Statist. Data Anal. 60, 12–31. doi:10.1016/j.csda.2012.10.021; arXiv:1203.3083*

$K \sim$ Poisson(1) given $K > 0$, with all parameters collapsed. The allocation sampler has four moves: add or remove an empty cluster, single-node Gibbs, M3, and absorb–eject. The add/remove and single-node moves "are sufficient to sample the space but have slow mixing". Ergodicity is asserted, not proved.

**B10 Model selection and clustering in stochastic block models based on the exact integrated complete data likelihood** [$K$ unknown] [context] Abs FT [core]

E. Côme, P. Latouche (2015). *Statistical Modelling 15(6), 564–589. doi:10.1177/1471082X15577017; arXiv:1303.2962*

Exact ICL, with Dirichlet and Beta priors integrated out, optimised greedily by node swaps from a large $K_{\text{up}}$. Swaps can empty clusters, and optional greedy merges follow. The paper reports that McDaid et al.'s collapsed sampler "tends to have poor mixing", that moves reducing complexity "are rarely accepted", and that $K$ "tends to be highly overestimated, even for a very large number of samples". In one example the sampler kept more than 240 clusters against about 80 for the greedy method. This is documented slowness in the *merging* direction.

**B11 Hierarchical clustering with discrete latent variable models and the integrated classification likelihood** [$K$ unknown] [context] Abs

E. Côme, N. Jouvin, P. Latouche, C. Bouveyron (2021). *Adv. Data Anal. Classif. 15(4), 957–986. doi:10.1007/s11634-021-00440-z; arXiv:2002.11577*

A hybrid genetic algorithm maximises ICL over partitions, which infers $K$. Greedy merges then build a hierarchy, with the Dirichlet parameter acting as a granularity regulariser (R package `greed`). There is no MCMC. Exactly which discrete latent variable models are covered was not confirmed.

**B12 Block clustering with collapsed latent block models** [$K$ random] [Q0] FT

J. Wyse, N. Friel (2012). *Statist. Comput. (2012)†; arXiv: 1011.2948*

A latent block model (biclustering) with truncated Poisson priors on the row and column cluster numbers, and a collapsed allocation sampler. The only diagnostics are integrated autocorrelation times. This is the bipartite analogue of P3.C.

**B13 Infinite-degree-corrected stochastic block model** [$K$ random] [Q0] FT

T. Herlau, M. N. Schmidt, M. Mørup (2014). *Phys. Rev. E 90(3), 032819. doi: 10.1103/PhysRevE.90.032819; arXiv: 1311.2520*

A CRP prior with degree correction. Gibbs on labels plus random-walk updates of hyperparameters. Runs of 1000 iterations "may exhibit mixing times significantly longer".

**B14 Probabilistic community detection with unknown number of communities** [$K$ random] [Q0] FT [core]

J. Geng, A. Bhattacharya, D. Pati (2019). *J. Amer. Statist. Assoc. 114(526), 893–905. doi: 10.1080/01621459.2018.1458618; arXiv: 1602.08062*

The MFM-SBM: $k \sim$ truncated Poisson(1), Dir($\gamma$) weights and Beta$(a, b)$ block probabilities. A Gibbs sampler integrates out $k$ and the weights through the MFM urn, draws the block probabilities from their Beta conditionals, and can open new clusters. The theory covers Bayes risk for known $K$, posterior concentration on the true clustering, and $\Pi(k = K \mid A) \geq 1 - e^{-cn^q}$ *for* $K \in \{2, 3\}$ *only*. General $K$ is "beyond the scope". Convergence of the sampler is shown empirically. This is the closest published sampler to P3.C's (S1), which integrates the block probabilities out instead. Under its MFM weights the opening cost carries an extra factor $n^{-\gamma}$ (Section 2.2).

**B15 Inference for a generalised stochastic block model with unknown number of blocks and non-conjugate edge models** [$K$ random] [Q0] FT

M. Ludkin (2020). *Comput. Statist. Data Anal. 152, 107051†. doi: 10.1016/j.csda.2020.107051; arXiv: 1909.09421*

$K - 1 \sim$ Poisson, with non-conjugate weighted edges. Reversible jump with split/merge and birth/death of empty blocks. "Assessing the convergence of a reversible jump Markov chain is non-trivial".

**B16 Bayesian estimation of the latent dimension and communities in stochastic blockmodels** [$K$ random] [Q0] FT

F. Sanna Passino, N. A. Heard (2020). *Statist. Comput. 30(5), 1291–1307. doi: 10.1007/s11222-020-09946-6; arXiv: 1904.05333*

Geometric prior on $K$ with Dir($\alpha/K$) weights, and a random latent dimension. Collapsed Gibbs, split–merge, create/remove empty cluster, and moves that change the dimension. Convergence is checked across chains.

**B17 Consistent Bayesian community detection** [$K$ random] [Q0] FT [core]

Z. Jiang, S. T. Tokdar (2021). *arXiv: 2101.06531; no journal version found*

$K \sim$ Poisson(1), with an assortative (diagonally dominant) prior on block probabilities. The posterior probability of $\{K = k_0\} \cap \{Z = Z_0\}$ tends to one for $k_0 \lesssim \sqrt{n}$ (Theorem 3.5), with a sparse extension. The sampler uses McDaid-type moves and simulations use $n \leq 75$. There is no MCMC theory.

**B18 Extended stochastic block models with application to criminal networks** [$K$ random] [Q0] FT

S. Legramanti, T. Rigon, D. Durante, D. B. Dunson (2022). *Ann. Appl. Statist. 16(4), 2369–2395. doi: 10.1214/21-AOAS1595; arXiv: 2007.08569*

Gibbs-type partition priors (Dirichlet-multinomial, DP, Pitman–Yor, MFM, Gnedin) with collapsed single-node Gibbs. Traceplots "suggest rapid convergence".

**B19 Recent Gibbs-type and partially exchangeable SBMs** [$K$ random] [Q0] Abs/FT/Sec

D. Durante, F. Gaffi, A. Lijoi, I. Prünster (2024); C. Lu, D. Durante, N. Friel (2024); B. Betancourt, E. Dilma, J. Sosa (2025) (2024–2025). *arXiv: 2410.10619; arXiv: 2410.23838; arXiv: 2505.19399*

- *Partially exchangeable SBMs for node-coloured multilayer networks.* Hierarchical random-partition priors and collapsed Gibbs.

- *Zero-inflated SBM.* A supervised Gnedin-process (MFM-type) prior. Single-node data-augmentation Gibbs in which the weight of a new group depends on the number of occupied groups. No split–merge; traceplots only. Read in full.
- *Hybrid Bayesian models.* DP, Pitman–Yor and Gnedin priors with Metropolis-within-Gibbs (secondary summary only).

None of the three makes a convergence statement.

**B20 Hierarchical stochastic block model for community detection in multiplex networks; Bayesian community detection for networks with covariates** [$K$ random] [Q0] Abs / FT

A. A. Amini et al.† (hierarchical SBM for multiplex networks); X. Shen, A. A. Amini, N. Josephs, L. Lin (2025) (2022–2025). *Bayesian Anal., advance publication, year†, doi: 10.1214/22-BA1355 (arXiv: 1904.05330); Bayesian Anal. 20(3), 735–762 (doi: 10.1214/24-BA1415)*

In the first, the number of communities in each layer is inferred, with labels tied by a hierarchical Dirichlet prior, using a slice sampler. In the second, a covariate-dependent product-partition prior is used with Gibbs. Neither has convergence theory.

**B21 Bayesian nonparametric community detection in assortative stochastic block models** [$K$ random] [Q0] FT

M. Amongero, P. De Blasi (2025). *arXiv: 2506.19576*

A Gnedin prior on $k$ with an assortativity cutoff. Gibbs sampling with Algorithm 8-type auxiliary blocks. There are no theorems, and convergence is checked by $\hat{R}$.

**B22 A Bayesian nonparametric stochastic block model for directed acyclic graphs** [$K$ random] [Q0] FT

C. Lee, M. Battiston (2025). *J. Comput. Graph. Statist. (2025)†, doi: 10.1080/10618600.2025.2554680; arXiv: 2301.07513*

A Pitman–Yor partition prior. The chain is "occasionally getting stuck at $K_n = 4$".

**B23 Bayesian degree-corrected SBMs; a Metropolis sampler for the SBM** [$K$ known]† [Q0] FT

L. Peng, L. Carvalho; Zhao, Ye & Huang (2016†, 2021). *arXiv: 1309.4796 (venue not checked); Entropy 23(1), 65 (doi: 10.3390/e23010065)*

In the first, $K$ is fixed and an uncollapsed Pólya–Gamma Gibbs sampler is used. In the second, the Metropolis run length is chosen empirically, and it is "unknown whether $O(n \log n)$ is sufficient"; how it treats $K$ was not recorded.

## 1.4 Description length, physics-style inference and resolution limits

**D1 Stochastic blockmodels and community structure in networks** [$K$ known] [context] Abs

B. Karrer, M. E. J. Newman (2011). *Phys. Rev. E 83, 016107. doi: 10.1103/PhysRevE.83.016107; arXiv: 1008.3926*

Introduces the degree-corrected SBM, used by most practical samplers (D9, D6). $K$ is fixed.

**D2 Asymptotic analysis of the stochastic block model for modular networks and its algorithmic applications** [context] FT

A. Decelle, F. Krzakala, C. Moore, L. Zdeborová (2011). *Phys. Rev. E 84, 066106; arXiv: 1109.3041*

Cavity-method phase diagram with belief propagation. Gibbs sampling equilibrates in linear time where belief propagation works, and the equilibration time is exponential in $N$ in the hard phase. Not rigorous.

**D3 Parsimonious module inference in large networks** [$K$ unknown] [Q0] Abs FT [core]

T. P. Peixoto (2013). *Phys. Rev. Lett. 110, 148701. doi: 10.1103/PhysRevLett.110.148701; arXiv: 1212.4794*

Minimum description length for SBMs and degree-corrected SBMs. "The maximum number of detectable blocks scales as $\sqrt{N}$" at fixed average degree ($B_{\max} \sim \sqrt{E}$ in general), with detectability condition $\langle k \rangle > 2 \ln B / I$.

- The limit comes from flat priors on the block matrix and the partition. It is a scaling argument, not a theorem.
- The inference is a multilevel Monte Carlo of cost $O(\tau N \log N)$, with the mixing time $\tau$ not analysed.

This is a statistical ceiling on $K$ that P3.C's growing-$K_0$ regime must respect.

**D4 Hierarchical block structures and high-resolution model selection in large networks** [$K$ unknown] [context] Abs [core]

T. P. Peixoto (2014). *Phys. Rev. X 4, 011047. doi: 10.1103/PhysRevX.4.011047; arXiv: 1310.4377*

The nested SBM. Hierarchical priors replace the $\sqrt{N}$ characteristic block size with one that "scales only logarithmically with $N$". That implies up to about $N/\log N$ blocks, although this is not stated verbatim. Hierarchies are exactly where invisible sub-community splits arise (Definition 2.1).

**D5 Efficient Monte Carlo and greedy heuristic for the inference of stochastic block models** [$K$ known] [Q1] FT

T. P. Peixoto (2014). *Phys. Rev. E 89, 012804; arXiv: 1310.4378*

Single-node Metropolis–Hastings with neighbour-informed proposals. The number of blocks is fixed in each run. "Fulfills the ergodicity condition".

**D6 Nonparametric Bayesian inference of the microcanonical SBM; Nonparametric weighted SBMs; Bayesian stochastic blockmodeling** [$K$ random] [Q1] FT

T. P. Peixoto (2017, 2018, 2019). *Phys. Rev. E 95, 012317; Phys. Rev. E 97, 012306; book chapter (arXiv: 1705.10225)*

Microcanonical and nested priors with $P(B) = 1/N$. Single-node Metropolis–Hastings moves, including into a new group. Ergodicity is argued informally. The chapter warns that asymptotics "may take a very long time to be realized".

**D7 Merge-split Markov chain Monte Carlo for community detection** [$K$ random] [Q1] FT [core]

T. P. Peixoto (2020). *Phys. Rev. E 102, 012305; arXiv: 2003.07070*

Merge–split moves with exactly computed reverse probabilities, via auxiliary staging splits. Mixing is "several orders of magnitude" faster, and single-node moves are effectively non-ergodic even on small networks. For the football network, the easiest single-node move has posterior ratio about $6 \times 10^{-8}$. All evidence is empirical.

**D8 Estimating the number of communities in a network** [$K$ random] [Q0] FT [core]

M. E. J. Newman, G. Reinert (2016). *Phys. Rev. Lett. 117, 078301; arXiv: 1605.02753*

A uniform prior on $k \leq n$ with empty groups allowed. Heat-bath node moves plus $k \pm 1$ moves, with an increase accepted with probability $k/(n+k)$. The chain "can get stuck in metastable states", so restarts are used.

**D9 Efficient method for estimating the number of communities in a network** [$K$ random] [Q1] FT [core]

M. A. Riolo, G. T. Cantwell, G. Reinert, M. E. J. Newman (2017). *Phys. Rev. E 96(3), 032310. doi: 10.1103/PhysRevE.96.032310; arXiv: 1706.02324*

A restaurant-type prior on $(g, k)$ for degree-corrected SBMs, with Metropolis node moves including moves to a new group. From random starts the sampler is accurate up to about 20 groups and *underestimates* $k$ beyond that. From the planted partition it recovers $k$. The authors call non-equilibration "the primary limitation". *This is the phenomenon P3.C sets out to explain.*

**D10 Model selection for degree-corrected block models** [$K$ known] [context] FT

X. Yan, C. R. Shalizi, J. E. Jensen, F. Krzakala, C. Moore, L. Zdeborová, P. Zhang, Y. Zhu (2014). *J. Stat. Mech. P05007. doi: 10.1088/1742-5468/2014/05/P05007; arXiv: 1207.3994*

A likelihood-ratio test of SBM against degree-corrected SBM using belief propagation. Classical $\chi^2$ theory fails in sparse graphs, so a corrected null is derived. AIC and DIC fail for the same reason. Estimating $K$ is deferred.

**D11 Cross-validation estimate of the number of clusters in a network** [$K$ unknown] [context] FT

T. Kawamoto, Y. Kabashima (2017). *Sci. Rep. 7, 3327. doi: 10.1038/s41598-017-03623-x; arXiv: 1605.07915*

Leave-one-out cross-validation criteria computed by belief propagation. The Bethe free energy and the Bayes-optimal criterion tend to overestimate $q$ on real networks. The Bayes-optimal criterion recovers the planted $q$ down to the detectability threshold, while the Gibbs criterion underfits.

**D12 Stochastic block models: a comparison of variants and inference methods** [context] Abs

T. Funke, T. Becker (2019). *PLoS ONE 14(4), e0215296. doi: 10.1371/journal.pone.0215296*

Puts SBM variants, including degree-corrected and hierarchical SBMs, into one notation and compares inference methods. Proposes a heuristic for the number of Metropolis–Hastings steps. How $K$ is handled was not confirmed.

## 1.5 Statistical theory: Bayesian posteriors and the number of communities

**S1 Bayesian community detection** [$K$ known] [context] Abs

S. L. van der Pas, A. W. van der Vaart (2018). *Bayesian Anal. 13(3), 767–796. doi:10.1214/17-BA1078; arXiv:1608.04242*

With $K$ known, the posterior mode is strongly consistent when the expected degree is $\gg (\log n)^2$.

**S2 A general framework for Bayes structured linear models** [$K$ unknown] [context] Abs FT (Section 5.1) [core]

C. Gao, A. W. van der Vaart, H. H. Zhou (2020). *Ann. Statist. 48(5)†. doi:10.1214/19-AOS1909; arXiv:1506.02174*

Covers SBMs with unknown $K$ (Section 5.1). The prior is $\Pi(k) \propto \Gamma(k^2)/\Gamma(k^2/2)\, e^{-D(k^2+n\log k)}$, uniform labels with no empty cluster, and an elliptical-Laplace prior on the connectivity matrix.

- Adaptive posterior contraction at the minimax rate $k_*^2 + n\log k_*$, "the first adaptive Bayes estimator for the SBM".
- Posterior model complexity is bounded by $(1+\delta)$ times the truth's. This is an upper bound, not consistency for $k$.
- Computation is left open, which is exactly where P3.C enters.

**S3 Optimal Bayesian estimation in stochastic block models; posterior contraction rates for SBMs** [$K$ known]† [context] Abs

D. Pati, A. Bhattacharya (2015); Ghosh, Pati, Bhattacharya (2020) (2015, 2020). *arXiv:1505.06794; Sankhyā A 82(2), 448–476*

Optimal posterior convergence rates with multinomial–Dirichlet labels. $K$ appears to be fixed. Whether the first is the preprint of the second was not checked.

**S4 Uncertainty quantification in the stochastic block model with an unknown number of classes** [$K$ random] [context] Abs [core]

J. van Waaij, B. J. K. Kleijn (2020). *arXiv:2005.01362*

A prior on the number of classes, which may grow with $n$ depending on sparsity. Explicit non-asymptotic contraction rates, credible sets that are confidence sets (after enlargement in the very sparse case), and consistent posterior-odds tests for the number of classes with explicit bounds on both error types. (Checked against the arXiv abstract for this document.)

**S5 Estimation of the number of communities in the stochastic block model** [$K$ unknown] [context] FT [core]

A. Cerqueira, F. Leonardi (2020). *IEEE Trans. Inf. Theory 66(10), 6403–6412. doi:10.1109/TIT.2020.3016331; arXiv:1804.03509*

A Krichevsky–Trofimov integrated likelihood, with Dirichlet$(\frac{1}{2})$ and Beta$(\frac{1}{2},\frac{1}{2})$ priors, minus the penalty $\left[\frac{k(k-1)(2k-1)}{12} + \frac{k(k-1)}{2} + (1+\epsilon)\frac{k-1}{2}\right]\log n$, maximised over $1 \le k \le n$. It is strongly consistent for $K$ in dense and sparse ($n\rho_n \to \infty$) regimes, with no upper bound on $K$ needed. This is a penalised evidence, not the plain posterior. It shows that the integrated likelihood carries the information. Whether the *unpenalised* posterior is consistent is target C2.

**S6 Consistent Bayesian community recovery in multilayer networks** [context] Abs

K. Alaluusua, L. Leskelä (2022). *Proc. IEEE ISIT 2022, 2726–2731. doi:10.1109/ISIT50566.2022.9834757; arXiv:2202.05823*

Conditions on parameter separation for posterior exact and almost-exact recovery in multilayer SBMs. How $K$ is treated was not checked.

**S7 Evidence thresholds of collapsed structured block models** [context] Abs

Papamichalis, Ruane (2026). *arXiv:2601.02828*

The summed evidence switches at the Kesten–Stigum threshold, while the best single partition keeps the planted structure only above a higher first-moment bound. This bears on which partition a collapsed sampler targets.

**S8 Consistency for the number of clusters in mixtures (context for C2)** [context] Abs [core]

J. W. Miller, M. T. Harrison (2014, 2018); J. W. Miller (2023); F. Ascolani, A. Lijoi, G. Rebaudo, G. Zanella (2023); Alamichel et al. (2024) (2014–2024). *J. Mach. Learn. Res. 15, 3333–3370; J. Amer. Statist. Assoc. 113(521), 340–356; Dependence Modeling 11(1), 1–9 (doi:10.1515/demo-2022-0150); Biometrika 110(2), 551–558; Scand. J. Statist. 51(4), 1619–1660 (doi:10.1111/sjos.12739)*

Checked against the abstracts for this document.

- Miller & Harrison (2014): the posterior on the number of components of DP and Pitman–Yor mixtures is not consistent when the data come from a finite mixture.
- Miller (2023): MFM models are consistent for the number of components, via Doob's theorem, at almost every parameter value.
- Ascolani et al. (2023): consistency is restored when the DP concentration parameter gets a prior.
- Alamichel et al. (2024) review (in)consistency.

P3.C's Target 2.7 is that SBMs differ. Summed over nodes, the odds of a one-node extra block are about $K_0\alpha(n/K_0)^{-K_0/2}$ in an SBM, but of order $\alpha$ in a fixed-dimensional mixture.

## 1.6 Frequentist estimation and testing of the number of communities

These entries come from the companion bibliography, where they were verified. They show that $K$ is statistically identifiable, and estimable in polynomial time, in the regimes P3.C uses.

**F1 Hypothesis testing for automated community detection in networks** [$K$ unknown] [context] Abs
P. J. Bickel, P. Sarkar (2016). *J. R. Stat. Soc. Ser. B 78(1), 253–273*
A Tracy–Widom test of $K = 1$ applied recursively by bipartitioning. This is a statistical analogue of spectral split proposals.

**F2 A goodness-of-fit test for stochastic block models** [$K$ unknown] [context] Abs
J. Lei (2016). *Ann. Statist. 44(1), 401–424*
A Tracy–Widom limit for the largest singular value of the standardised residual matrix. Sequential tests give a consistent estimate of $K$.

**F3 Likelihood-based model selection for stochastic block models** [$K$ unknown] [context] Abs
Y. X. R. Wang, P. J. Bickel (2017). *Ann. Statist. 45(2), 500–528*
Asymptotics of log-likelihood ratios between SBMs with different $K$, and a consistent penalised criterion. It is the frequentist counterpart of the Occam factor in P3.C.

**F4 Network cross-validation for determining the number of communities in network data** [$K$ unknown] [context] Abs
K. Chen, J. Lei (2018). *J. Amer. Statist. Assoc. 113(521), 241–251*
Node-pair cross-validation for SBMs and degree-corrected SBMs, consistent against underestimation.

**F5 Network cross-validation by edge sampling** [$K$ unknown] [context] Abs
T. Li, E. Levina, J. Zhu (2020). *Biometrika 107(2), 257–276*
Edge cross-validation with low-rank completion. Consistent selection of $K$ and of the model.

**F6 Corrected Bayesian information criterion for stochastic block models** [$K$ unknown] [context] Abs FT
J. Hu, H. Qin, T. Yan, Y. Zhao (2020). *J. Amer. Statist. Assoc. 115(532), 1771–1783; arXiv: 1611.01238*
CBIC equals the profile log-likelihood minus $\lambda n \log k$ minus $\frac{1}{2}k(k+1)\log n$, where the $\lambda n \log k$ term comes from a prior on labels. It is consistent, while plain BIC tends to overestimate $k$.

**F7 Using maximum entry-wise deviation to test the goodness of fit for stochastic block models** [$K$ unknown] [context] Abs
J. Hu, J. Zhang, H. Qin, T. Yan, J. Zhu (2021). *J. Amer. Statist. Assoc. 116(535), 1373–1382*
An extreme-value goodness-of-fit test. Sequential testing estimates $K$.

**F8 A unified framework for spectral clustering in sparse graphs** [$K$ unknown] [context] Abs
L. Dall'Amico, R. Couillet, N. Tremblay (2021). *J. Mach. Learn. Res. 22(217)*
A regularised Laplacian related to the Bethe Hessian. It estimates $K$ and the communities in sparse, heterogeneous graphs.

**F9 Hierarchical community detection by recursive partitioning** [$K$ unknown] [context] Abs [core]
T. Li, L. Lei, S. Bhattacharyya, K. Van den Berge, P. Sarkar, P. J. Bickel, E. Levina (2022). *J. Amer. Statist. Assoc. 117(538), 951–968*
Recursive bipartitioning by the sign of an eigenvector, with a stopping rule, recovers a binary-tree SBM hierarchy. It needs weaker conditions at the top levels than $K$-way methods. This is the closest analogue of the spectral split in (S2), and the binary-tree SBM is a natural home for invisible sub-community splits.

**F10 Optimal estimation of the number of network communities** [$K$ unknown] [context] Abs

J. Jin, Z. T. Ke, S. Luo, M. Wang (2023). *J. Amer. Statist. Assoc. 118(543), 2101–2116*

StGoF steps up in $K$ with an $N(0,1)$ goodness-of-fit statistic. It is consistent and optimal under severe degree heterogeneity.

**F11 Universal rank inference via residual subsampling; adjusted chi-square test for degree-corrected block models** [$K$ unknown] [context] Abs

X. Han, Q. Yang, Y. Fan (2023); L. Zhang, A. A. Amini (2023) (2023). *Ann. Statist. 51(3), 1109–1133; Ann. Statist. 51(6), 2366–2385*

Two sequential tests that estimate $K$ consistently: a subsampled residual rank test with a normal limit, and an adjusted $\chi^2$ test for degree-corrected SBMs.

**F12 On the estimation of the number of communities for sparse networks** [$K$ unknown] [context] Abs

N. Hwang, J. Xu, S. Chatterjee, S. Bhattacharyya (2024). *J. Amer. Statist. Assoc. 119(547), 1895–1910*

Counting the negative eigenvalues of a tuned Bethe Hessian is consistent in sparse SBMs and degree-corrected SBMs.

**F13 Testing for latent structure via the Wilcoxon–Wigner random matrix of normalized rank statistics** [$K$ unknown] [context] Abs

J. Z. Liao, J. Cape (2026). *Biometrika 113(2), asag003*

Distribution-free spectral tests for community structure.

## 1.7 Thresholds with many or growing numbers of communities

**T1 Spectral clustering and the high-dimensional stochastic blockmodel** [context] FT

K. Rohe, S. Chatterjee, B. Yu (2011). *Ann. Statist. 39(4). doi: 10.1214/11-AOS887*

The first polynomial-time guarantee with growing $K$. In their example, $k = O(n^{1/4}/\log n)$ gives a misclustered fraction of $o(n^{-1/4})$.

**T2 Stochastic blockmodels with a growing number of classes** [context] FT

D. S. Choi, P. J. Wolfe, E. M. Airoldi (2012). *Biometrika 99(2), 273–284. doi: 10.1093/biomet/asr053*

The MLE misclassifies $o_P(N)$ nodes if $K = O(N^{1/2})$, the average degree is $\gg (\log N)^{3+\delta}$, and class sizes are $\Omega(N/K)$.

**T3 Exact recovery in the stochastic block model** [context] Abs [core]

E. Abbe, A. S. Bandeira, G. Hall (2016). *IEEE Trans. Inf. Theory 62(1), 471–487. doi: 10.1109/TIT.2015.2490670; arXiv: 1405.3267*

Two communities with $p = \alpha \log n/n$ and $q = \beta \log n/n$: exact recovery is possible if and only if $(\alpha+\beta)/2 - \sqrt{\alpha\beta} > 1$. An SDP works near the threshold.

**T4 Community detection in general stochastic block models: fundamental limits and efficient algorithms for recovery** [context] FT [core]

E. Abbe, C. Sandon (2015). *FOCS 2015, 670–688. doi: 10.1109/FOCS.2015.47; arXiv: 1503.00609*

For fixed $k$ and probabilities $Q \log n/n$, exact recovery is possible if and only if a Chernoff–Hellinger divergence between all pairs of columns is at least one. It is achieved in quasi-linear time. For $K_0$ equal communities in the planted partition model this is $(\sqrt{a} - \sqrt{b})^2 > K_0$. There is no computational gap for exact recovery at fixed $k$.

**T5 Proof of the achievability conjectures for the general stochastic block model** [context] Abs

E. Abbe, C. Sandon (2018). *Comm. Pure Appl. Math. 71(7), 1334–1406. doi: 10.1002/cpa.21719; arXiv: 1512.09080*

Acyclic belief propagation reaches the Kesten–Stigum bound for all $k$. An inefficient "typical clustering" sampler beats Kesten–Stigum from $k = 4$ on, roughly by a factor $k/\ln k$.

**T6 Community detection and stochastic block models: recent developments** [context] Abs

E. Abbe (2018). *J. Mach. Learn. Res. 18(177), 1–86; arXiv: 1703.10146*

The standard survey of thresholds, gaps and algorithms.

**T7 Consistency thresholds for the planted bisection model** [context] FT [core]

E. Mossel, J. Neeman, A. Sly (2016). *Electron. J. Probab. 21, paper 21, 1–24. doi: 10.1214/16-EJP4185 (and STOC 2015)*

Sharp conditions for strong and weak consistency. Exact recovery is possible if and only if, with high

probability, every vertex has more neighbours in its own community than in the other. That is exactly the local stability of the truth under single-site moves, the starting point for target C3.

**T8 Minimax rates of community detection in stochastic block models** [context] FT

A. Y. Zhang, H. H. Zhou (2016). *Ann. Statist. 44(5), 2252–2280*†. *doi:10.1214/15-AOS1428*

The minimax misclassification proportion is $\exp(-(1+o(1))nI/(\beta K))$ for $K \geq 3$, and at least a constant if $nI/K = O(1)$. This is the statistical benchmark with many groups.

**T9 Achieving optimal misclassification proportion in stochastic block models** [context] FT [core]

C. Gao, Z. Ma, A. Y. Zhang, H. H. Zhou (2017). *J. Mach. Learn. Res. 18(60), 1–45*

Spectral initialisation followed by node-wise refinement attains the minimax rate. Refinement needs $(a-b)^2/(ak\log k) \to \infty$, and spectral initialisation needs $(a-b)^2/(ak^3\log k) \to \infty$. In P3.C's regimes these polynomial-time methods succeed.

**T10 Stochastic block model and community detection in sparse graphs: a spectral algorithm with optimal rate of recovery** [context] FT

P. Chin, A. Rao, V. Vu (2015). *Proc. COLT, PMLR 40, 391–423*

Spectral recovery with optimal rate for two communities. With $k$ communities the condition carries a $k^2$ factor.

**T11 Statistical-computational tradeoffs in planted problems and submatrix localization with a growing number of clusters and submatrices** [context] FT

Y. Chen, J. Xu (2016). *J. Mach. Learn. Res. 17(27), 1–57; arXiv:1402.1267*

For $r$ clusters of size $K$:

- the MLE succeeds once $K(p-q)^2 \gtrsim q(1-q)\log n$;
- convex relaxation needs $K^2(p-q)^2 \gtrsim pK\log n + q(1-q)n$;
- the gap is conjectured intrinsic.

**T12 Information-theoretic thresholds for community detection in sparse networks** [context] Abs [core]

J. Banks, C. Moore, J. Neeman, P. Netrapalli (2016). *Proc. COLT, PMLR 49, 383–416; arXiv:1601.02658*

Detection is information-theoretically possible at average degree $\Theta(\log q/(q\lambda^2))$, against the Kesten–Stigum value $1/\lambda^2$. This beats Kesten–Stigum for $q \geq 5$ (disassortative) and $q \geq 11$ (assortative): the gap grows like $q/\log q$.

**T13 Efficient Bayesian estimation from few samples: community detection and related problems** [context] FT

S. B. Hopkins, D. Steurer (2017). *FOCS 2017, 379–390. doi:10.1109/FOCS.2017.42*

Low-degree evidence for the Kesten–Stigum barrier: below Kesten–Stigum, polynomials with non-trivial correlation need degree at least $n^c$.

**T14 Sharp phase transitions in estimation with low-degree polynomials; Exact phase transitions for SBMs and reconstruction on trees** [context] FT/Abs

Y. Sohn, A. S. Wein (2025); E. Mossel, A. Sly, Y. Sohn (2022) (2022, 2025). *STOC 2025, 891–902 (doi:10.1145/3717823.3718294; arXiv:2502.14407); arXiv:2212.03362*

Low-degree polynomials have a sharp recovery barrier at Kesten–Stigum. Kesten–Stigum is the information-theoretic threshold for $q = 3, 4$ at large degree (in the antiferromagnetic case $q = 4$ the answer depends on the degree) but not for $q \geq 5$.

**T15 Stochastic block models with many communities and the Kesten–Stigum bound** [context] FT [core]

Chin†, E. Mossel, Y. Sohn, A. S. Wein (2025). *Proc. COLT, PMLR 291, 1253–1258 (extended abstract); arXiv:2503.03047*

For a growing number of communities $q$:

- non-backtracking walks succeed above Kesten–Stigum if $n/(q\log^3 n) \to \infty$;
- low-degree polynomials fail below Kesten–Stigum for $q \leq n^{1/2-\epsilon}$;
- for $q = n^\chi$ with $\chi > \frac{1}{2}$, efficient recovery is possible below Kesten–Stigum;
- information-theoretically, recovery is possible if $d\lambda > 14$ and impossible if $d\lambda < 1$.

This is the reference picture for growing $K_0$. P3.C stays far above these thresholds.

**T16 Cavity-method thresholds and phase types** [context] FT/Abs

A. Coja-Oghlan, F. Krzakala, W. Perkins, L. Zdeborová (2018); L. Zdeborová, F. Krzakala (2016); F. Ricci-Tersenghi,

G. Semerjian, L. Zdeborová (2019); F. Antenucci, S. Franz, P. Urbani, L. Zdeborová (2019) (2016–2019). *Adv. Math. 333, 694–795 (doi: 10.1016/j.aim.2018.05.029); Adv. Phys. 65(5), 453–552; Phys. Rev. E 99(4), 042109; Phys. Rev. X 9, 011020*

- A rigorous replica-symmetric formula, with the information-theoretic threshold below Kesten–Stigum for $q \geq 5$ in the disassortative SBM.
- The physics review of thresholds.
- A typology of transitions, including a "hybrid-hard" phase.
- The glassy nature of the hard phase: exponentially many metastable states.

These concern the hard phase, which P3.C avoids.

## 1.8 Mixing, hardness and metastability: rigorous templates

**M1 Mixing time of Metropolis–Hastings for Bayesian community detection** [$K$ known] [Q3] FT [core]
B. Zhuo, C. Gao (2021). *J. Mach. Learn. Res. 22(10), 1–89 (paper 18-770); arXiv: 1811.02612*

The only rapid-mixing theorem for a Bayesian SBM posterior.

- *Setting.* Homogeneous SBM with $K$ known. Uniform prior on balanced labellings and Beta priors, with a tempered posterior $\Pi^{\xi}$, $\xi \geq 1$. Single-node Metropolis–Hastings restricted to balanced partitions.
- *Conditions.* Above the exact-recovery threshold and from a warm start.
- *Result.* $\tau_\epsilon \lesssim n^2(n^2 I + \log \epsilon^{-1})$, proved by canonical paths.

The authors find the restriction to balanced partitions "seems to be necessary". Target C3 extends this theorem to random $K$, and its posterior-ratio lemmas are the starting point for C1.

**M2 The Metropolis algorithm for graph bisection** [$K$ known] [Q3] Abs
M. Jerrum, G. B. Sorkin (1998). *Discrete Appl. Math. 82†. doi: 10.1016/S0166-218X(97)00133-9*

Fixed-temperature Metropolis finds a planted bisection in $O(n^{2+\epsilon})$ steps when the gap is large. This is a hitting-time result.

**M3 Large cliques elude the Metropolis process** [Q3] Meta Sec [core]
M. Jerrum (1992). *Random Struct. Alg. 3(4), 347–359. doi: 10.1002/rsa.3240030402*

In $G(n, \frac{1}{2})$ with a planted clique of size $n^\alpha$, $\alpha < \frac{1}{2}$, there is a start from which Metropolis needs superpolynomial time, restated as $n^{\Omega(\log n)}$, to find a clique of size $(1+\epsilon)\log_2 n$. This is the classical "gateway" bottleneck: a local chain must pass through unlikely intermediate states to build planted structure, as a new community must in P3.C.

**M4 Planted cliques elude Metropolis; overlap gaps; free-energy wells** [Q3] Abs
Z. Chen, E. Mossel, I. Zadik (2023); D. Gamarnik, I. Zadik (2024); G. Ben Arous, A. S. Wein, I. Zadik (2020) (2020–2024). *SODA 2023† and arXiv: 2204.01911; Ann. Appl. Probab. 34(4), 3375–3434; Proc. COLT, PMLR 125, 479–482*

- Metropolis fails for planted cliques of size $n^\alpha$, any $\alpha < 1$, even from the empty clique; simulated tempering also fails.
- Overlap gaps for planted clique.
- Free-energy wells imply exponential hitting times for local MCMC in sparse PCA.

These are the modern versions of the bottleneck argument.

**M5 Glauber dynamics for the mean-field Potts model** [Q3] Abs [core]
P. Cuff, J. Ding, O. Louidor, E. Lubetzky, Y. Peres, A. Sly (2012). *J. Stat. Phys. 149(3), 432–477. doi: 10.1007/s10955-012-0599-2; arXiv: 1204.4503*

For $q \geq 3$:

- mixing in about $n \log n$ with cutoff below the spinodal point $\beta_s(q)$;
- $\Theta(n^{4/3})$ at $\beta_s$;
- exponential in $n$ above it.

This is the rigorous first-order picture. A posterior that holds two communities in one block is analogous to the metastable phase.

**M6 Swendsen–Wang dynamics on the mean-field Potts model** [Q3] Abs

V. K. Gore, M. R. Jerrum (1999); A. Galanis, D. Štefankovič, E. Vigoda (2019); R. Gheissari, E. Lubetzky, Y. Peres (2020) (1999–2020). *J. Stat. Phys. 97(1–2), 67–86; Random Struct. Alg. 54(1), 82–147 (arXiv: 1502.06593); Ann. Inst. H. Poincaré Probab. Stat. 56(1), 68–86 (arXiv: 1702.05797)*

Cluster (global) moves also fail at first-order transitions: exponential mixing at or near criticality for $q \geq 3$, with a full phase diagram for Swendsen–Wang. *Caution for P3.C:* non-local moves are not automatically fast. The split–merge upper bound must use the specific structure, namely that splitting along the truth increases the posterior.

**M7 Metastability of the Potts ferromagnet on random regular graphs** [Q3] FT [core]

A. Coja-Oghlan, A. Galanis, L. A. Goldberg, J. B. Ravelomanana, D. Štefankovič, E. Vigoda (2023). *Commun. Math. Phys. 401(1), 185–225. doi: 10.1007/s00220-023-04644-6; arXiv: 2202.05777*

On random regular graphs, the disordered and ordered phases are metastable over explicit ranges of $\beta$, with escape times $e^{\Omega(n)}$. Glauber dynamics and Swendsen–Wang mix in time $e^{\Omega(n)}$ over explicit ranges. The proof uses a planted-model conductance argument that can be adapted to P3.C's lower bounds.

**M8 Planting and MCMC sampling from the Potts model** [Q3] FT

A. Galanis, L. A. Goldberg, P. Smolarova (2024). *arXiv: 2410.14409 (venue not checked)*

Random-cluster dynamics on random regular graphs mixes in $O(n \log n \log(1/\epsilon))$ from a suitable all-in or all-out start. A good initialisation avoids metastability, which is the analogue of P3.C's over-clustered starts.

**M9 Exact thresholds for Ising–Gibbs samplers on general graphs** [Q3] Abs

E. Mossel, A. Sly (2013). *Ann. Probab. 41(1), 294–328. doi: 10.1214/11-AOP737; arXiv: 0903.2906*

If $(d-1)\tanh\beta < 1$, the Gibbs sampler mixes in $O(n \log n)$ on any graph of maximum degree $d$. Sharp thresholds on $G(n, d/n)$. This marks the weak-signal regime where single-site dynamics is fast.

**M10 Torpid mixing of tempering on the Potts model; conditions for rapid and torpid tempering** [Q3] FT / Abs

N. Bhatnagar, D. Randall (2004, 2016); D. B. Woodard, S. C. Schmidler, M. Huber (2009) (2004–2016). *SODA 2004; J. Stat. Phys. 164(3), 495–530 (doi: 10.1007/s10955-016-1526-8); Ann. Appl. Probab. 19(2), 617–640; Electron. J. Probab. 14(29), 780–804*

Tempering needs at least $e^{cn}$ steps at the first-order point of the 3-state mean-field Potts model, for any polynomial schedule. "Entropy-dampening" interpolants restore polynomial time. Conditions for rapid and torpid tempering follow. So tempering is not a remedy for P3.C's barrier, as P3.F argues for mixtures.

**M11 The stochastic block model has the overlap graph [sic] property for modularity** [$K$ known] [Q3] FT

S. Bhamidi, D. Gamarnik, R. van der Hofstad, N. Litvak, P. Prałat, F. Skerman, M. Tousinejad (2026). *arXiv: 2605.10911; ICALP 2026, LIPIcs 374, 28:1–28:20†*

An overlap gap for near-optimal-modularity partitions of an SBM with $k \geq 3$ blocks. Single-vertex heat-bath dynamics for a modularity-tilted measure needs $e^{cn}$ steps from the far set. This concerns a modularity measure, not the Bayesian posterior.

**M12 Glauber dynamics for two-community inference** [$K$ known] [Q3] FT / Abs

K. Liu, S. Mohanty, P. Raghavendra, A. Rajaraman, D. X. Wu (2024); K. Liu, S. Mohanty, A. Rajaraman, D. X. Wu (2024); A. Rajaraman, D. X. Wu (2025); A. S. Bandeira, A. El Alaoui, A. Rödder (2026) (2024–2026). *arXiv: 2405.20849; arXiv: 2405.06616; arXiv: 2512.02384; arXiv: 2607.06813*

- Locally stationary distributions: weak recovery by Glauber dynamics at a random time, for surrogate measures.
- Fast mixing in sparse random Ising models.
- Glauber dynamics tracks approximate message passing from a warm start.
- Polynomial mixing on high-overlap Gibbs measures, including $\mathbb{Z}_2$ synchronisation with side information.

All have two communities and $K$ known.

**M13 Theoretical and computational guarantees of mean field variational inference for community detection** [$K$ known] [context] Abs

A. Y. Zhang, H. H. Zhou (2020). *Ann. Statist. 48(5), 2575–2598†; arXiv: 1710.11268*

Batch coordinate-ascent variational inference has a linear convergence rate and reaches the minimax rate within $\log n$ iterations (the abstract does not state the initialisation condition). An algorithmic analogue of local mixing.

**M14 Lower bounds for the Gibbs sampler over mixtures of Gaussians** [$K$ known] [Q3] FT

C. Tosh, S. Dasgupta (2014). *Proc. ICML, PMLR 32(2), 1467–1475*

Conductance lower bounds for collapsed Gibbs on Gaussian mixtures: exponential in the separation when $k$ is misspecified, and of order $n^{\Omega(\alpha)}$ for the correct $k$. This is the mixture analogue of target C1.

## 1.9 Samplers that change $K$, and split–merge designs

**X1 Markov chain sampling methods for DP mixture models; split–merge samplers for conjugate and non-conjugate DP mixtures** [$K$ random] [Q1] FT/Abs

R. M. Neal (2000); S. Jain, R. M. Neal (2004, 2007) (2000–2007). *J. Comput. Graph. Statist. 9(2), 249–265; J. Comput. Graph. Statist. 13(1)†, 158–182 (doi: 10.1198/1061860043001); Bayesian Anal. 2(3), 445–472† (doi: 10.1214/07-BA219)*

Algorithm 3 is the template for (S1) on SBMs. Split–merge with restricted-Gibbs proposals is the template for (S2). Ergodicity is asserted for some algorithms; there are no rates.

**X2 Mixture models with a prior on the number of components** [$K$ random] [Q1] FT

J. W. Miller, M. T. Harrison (2018). *J. Amer. Statist. Assoc. 113(521), 340–356*

MFM partition priors with weights $n_c + \gamma$ and $\gamma V_n(t+1)/V_n(t)$, and the corresponding Algorithm 3. These are the MFM weights used in (S1). The partition sampler is finite-state.

**X3 Further split–merge designs** [$K$ random] [Q0]/[Q1] Meta/Abs/FT

D. B. Dahl, S. Newcomb (2022); J. Chang, J. W. Fisher III (2013); A. Bouchard-Côté, A. Doucet, A. Roth (2017) (2013–2022). *J. Statist. Comput. Simul. (doi: 10.1080/00949655.2021.1998502); NIPS 26; J. Mach. Learn. Res. 18(28)*

Sequentially allocated merge–split, sub-cluster splits that pair a non-ergodic restricted Gibbs step with split–merge, and particle-Gibbs split–merge. All have invariance or irreducibility only.

**X4 A fast non-reversible sampler for Bayesian (finite) mixture models** [$K$ known] [Q3] FT

F. Ascolani, G. Zanella; F. Ascolani, P. Manildo, G. Zanella (2025, 2026). *arXiv: 2510.03226 (v1 2025, v2 2026)*

Comparison theorems and prior-case scaling limits ($n^2$ versus $n$) for collapsed and non-reversible samplers with fixed $K$. v2 announces a random-partition variance comparison that was not seen. This is the closest rigorous analysis of collapsed clustering samplers, and it says nothing about creating clusters.

**X5 Geometric ergodicity of trans-dimensional Markov chain Monte Carlo algorithms** [$K$ random] [Q2] FT

Q. Qin (2025). *J. Amer. Statist. Assoc. 120(551), 1868–1878. doi: 10.1080/01621459.2024.2427432*

Spectral-gap decompositions into within-model and between-model parts. For P3.C, the "models" are the values of $K$, and the between-model gap is what the opening cost destroys.

**X6 Coupled partition samplers; component-first moves; telescoping sampling** [$K$ random] [Q1]/[Q2] FT

T. D. Nguyen, B. L. Trippe, T. Broderick (2022); M. E. J. Newman (2025); S. Frühwirth-Schnatter, G. Malsiner-Walli, B. Grün (2021) (2021–2025). *AISTATS, PMLR 151, 3483–3514; Statist. Comput., doi: 10.1007/s11222-025-10753-0; Bayesian Anal. 16(4)*

Optimal-transport couplings with meeting-time bounds that degenerate in $n$; a rejection-free sampler that picks a component first, whose fast removal of components is argued heuristically; and exact updates of $K$ given the partition. Each is a candidate design to test against the barrier in P3.C.

## 1.10 General Markov chain tools

**G1 Approximate counting, uniform generation and rapidly mixing Markov chains** [tool] Meta

A. Sinclair, M. Jerrum (1989). *Inform. and Comput. 82(1), 93–133†*

Conductance bounds for the spectral gap. Small conductance of a set implies slow mixing. This is the basic lower-bound tool.

**G2 Bounds on the $L^2$ spectrum for Markov chains and Markov processes: a generalization of Cheeger's inequality** [tool] Meta

G. F. Lawler, A. D. Sokal (1988). *Trans. Amer. Math. Soc. 309(2), 557–580*

Cheeger's inequality for Markov chains, $\Phi^2/8 \le \text{gap} \le 2\Phi$ in the reversible case†; the constant $\frac{1}{8}$ was later improved to $\frac{1}{2}$ by Jerrum and Sinclair†.

**G3 Geometric bounds for eigenvalues of Markov chains; Improved bounds for mixing rates of Markov chains and multicommodity flow** [tool] Meta

P. Diaconis, D. Stroock (1991); A. Sinclair (1992) (1991, 1992). *Ann. Appl. Probab. 1(1), 36–61†; Combin. Probab. Comput. 1(4), 351–370†*

Canonical paths and multicommodity flows, the upper-bound tools used by Zhuo & Gao (M1) and by Yang, Wainwright & Jordan for variable selection.

**G4 Mixing times are hitting times of large sets; Mixing and hitting times for finite Markov chains; Markov Chains and Mixing Times (2nd ed.)** [tool] Meta

Y. Peres, P. Sousi (2015); R. I. Oliveira (2012); D. A. Levin, Y. Peres, E. L. Wilmer (2017) (2012–2017). *J. Theoret. Probab. 28(2) (doi: 10.1007/s10959-013-0497-9); arXiv: 1108.1708; American Mathematical Society*

For lazy reversible chains, mixing times equal hitting times of large sets up to constants. With posterior concentration this reduces P3.C's mixing statements to hitting times of the truth. The textbook collects conductance, coupling and path methods.

**G5 Entrywise eigenvector analysis of random matrices with low expected rank; The rotation of eigenvectors by a perturbation III** [tool] Abs/Meta

E. Abbe, J. Fan, K. Wang, Y. Zhong (2020); C. Davis, W. M. Kahan (1970) (1970, 2020). *Ann. Statist. 48(3), 1452–1474; SIAM J. Numer. Anal. 7(1), 1–46†*

Entrywise ($\ell_\infty$) and subspace perturbation bounds for eigenvectors. These are the tools for proving that a spectral split of a merged block separates the communities, with $o(1)$ error fixed by single-site sweeps.

**G6 Markov chains for exploring posterior distributions; General state space Markov chains and MCMC algorithms** [tool] Sec/FT

L. Tierney (1994); G. O. Roberts, J. S. Rosenthal (2004) (1994, 2004). *Ann. Statist. 22(4), 1701–1762; Probab. Surveys 1, 20–71*

Qualitative convergence theory. For finite partition spaces it gives uniform ergodicity once the chain is irreducible and aperiodic.

## 1.11 What could not be verified

- *Bibliographic details marked †.*
  - Issue or pages of Daudin et al., Latouche et al., Gao, van der Vaart & Zhou, Jain & Neal, and Jerrum & Sorkin.
  - Venues of Peng & Carvalho, Galanis et al. (2024) and Lee & Battiston.
  - The co-authors of the hierarchical multiplex SBM paper.
  - The ICALP version of Bhamidi et al.
  - Page ranges of the classical tools (Section 1.10), which are given from standard citations.
  - The volume and number of Hofman & Wiggins, and the page numbers of Zhang & Zhou (2016, 2020), which come from secondary sources.
- *Provenance.* The entries of Section 1.6 were verified from their abstracts for the companion survey of theoretical clustering papers (2020–2026), not again for this document. Entries M13, X1, X3, T12, S4 and S8 were checked against abstracts or publisher pages during the final check of this document.
- *Content not checked.*
  - Daudin et al. (2008).
  - How Funke & Becker handle $K$.
  - Which models Côme et al. (2021) cover.
  - The algorithm section of Peixoto (2014, PRX), and whether "$N/\log N$" is stated verbatim.
  - The exact theorem wording of Jerrum (1992), which comes from a secondary source.
- *Not found despite targeted searches.*
  - Any mixing or hitting-time result for an SBM sampler with unknown $K$.

- Any consistency result for $K$ under a Dirichlet-process SBM posterior.
- Any analysis of moves that open communities.

# 2 Project P3.C in detail

## 2.1 Summary

**Working title.** *Why single-site samplers underestimate the number of communities: the cost of opening a community in Bayesian stochastic block models.*

**One-paragraph abstract (draft).** Bayesian SBMs with a prior on the number of communities are fitted by samplers that move one node at a time and can open a new community. We show that, under a Dirichlet-process prior and with $K$ blocks present, opening a community from a single node costs a factor of about $n^{-(K+2)/2}$ in posterior probability. The cost comes from the $K$ new edge probabilities that a new block requires, each estimated from about $n/K$ trials, and from the partition prior's factor $1/n$. When two communities have the same size and internal density and connect to the rest of the network in the same way, nothing offsets this cost. As a result, single-site samplers need asymptotically about $n^{(K_0+1)/2}$ updates to separate two merged communities, a polynomial whose degree grows with the number of communities $K_0$, and superpolynomial time when $K_0$ grows. Started from a single block, they discover only a bounded number of communities in polynomial time, which explains the underestimation reported in practice. In contrast, the posterior concentrates on the truth, and split–merge moves recover it in polynomial time. Under a mixture-of-finite-mixtures prior with parameter $\gamma$, all exponents increase by $\gamma$.

**The main targets** (Section 2.5):

- C1.1, the exponent law;
- C1.2, superpolynomial barriers when $K_0$ grows;
- C1.3, the underestimation law;
- C1.4, the visible–invisible dichotomy;
- C1.5, polynomial split–merge;
- C2.1–C2.2, posterior concentration and consistency of Dirichlet-process SBMs for $K$;
- C3, local mixing with random $K$;
- C4, the opening-cost principle.

## 2.2 Model, posterior and samplers

**Data and model.** $A \in \{0,1\}^{n\times n}$ is symmetric with zero diagonal. Given a partition $C$ of $[n]$ into blocks $c_1, \dots, c_K$ and a symmetric matrix $B \in [0,1]^{K\times K}$, the entries $A_{ij}$, $i < j$, are independent Bernoulli$(B_{kl})$ for $i \in c_k$ and $j \in c_l$. For blocks $k, l$ write:

- $E_{kl}$ for the number of edges between them, or within $c_k$ if $k = l$;
- $N_{kl}$ for the number of node pairs, $n_k n_l$ if $k \neq l$ and $n_k(n_k-1)/2$ if $k = l$.

**Prior and collapsed posterior.** The partition prior is DP($\alpha$), $p(C) = \alpha^K \prod_k (n_k-1)!/\alpha^{(n)}$, or MFM, $p(C) = V_n(K)\prod_k \gamma^{(n_k)}$. Given $C$, the entries $B_{kl}$, $k \le l$, are independent Beta$(a,b)$. Integrating $B$ out gives

$$\Pi_n(C) \;\propto\; p(C)\prod_{k\le l}\frac{\mathrm{B}(a+E_{kl},\, b+N_{kl}-E_{kl})}{\mathrm{B}(a,b)}, \tag{1}$$

with B the Beta function. This is the infinite relational model of B3 under the DP, and the MFM-SBM of B14 under the MFM (whose sampler draws $B$ from its conditionals instead of integrating it out). The state space is finite and every sampler below is irreducible and aperiodic, hence uniformly ergodic, so only rates are at stake.

**Single-site sampler (S1).** Remove node $i$ from its block, and let $d_{il}$ be the number of edges from $i$ to block $l$. Then:

- *Joining block $k$* has weight

$$w_k \prod_l \frac{\mathrm{B}(a+E_{kl}+d_{il},\, b+N_{kl}+n_l^{(k)}-E_{kl}-d_{il})}{\mathrm{B}(a+E_{kl},\, b+N_{kl}-E_{kl})},$$

  where $n_l^{(k)} = n_l$ is the number of new trials.

- *Opening a new block* has weight
$$w_{\text{new}} \prod_l \frac{\mathrm{B}(a + d_{il},\, b + n_l - d_{il})}{\mathrm{B}(a,b)}.$$
- *Weights.* Under the DP, $w_k = n_k$ and $w_{\text{new}} = \alpha$. Under the MFM, $w_k = n_k + \gamma$ and $w_{\text{new}} = \gamma V_n(K+1)/V_n(K)$. Since $V_n(K+1)/V_n(K)$ scales like $n^{-\gamma}$, the ratio $w_{\text{new}}/w_k$ is of order $n^{-1}$ under the DP but $n^{-1-\gamma}$ under the MFM.

Scans are random or systematic.

**Split–merge sampler (S2).** Sweeps of (S1) alternate with Metropolis–Hastings proposals. (For the theory, a lazy random mixture of the two kernels is preferable, since it is reversible; see Target 2.5.)

- *Split* (probability $\frac{1}{2}$): pick a block with at least four nodes uniformly at random. Split it by the sign of the leading eigenvector of its centred adjacency submatrix.
- *Merge* (probability $\frac{1}{2}$): pick two blocks uniformly at random. The merge is possible only if the spectral split of their union returns exactly that pair; otherwise it is rejected.
- *Acceptance.* The standard Metropolis–Hastings ratio, with proposal probabilities $\frac{1}{2}/\#\{\text{eligible blocks}\}$ and $\frac{1}{2}/\binom{K}{2}$.

This makes a valid, analysable kernel. Jain–Neal restricted-Gibbs proposals (X1) and Peixoto's merge–split (D7) are the practical variants.

**Truth, regimes and structural assumption.** The true partition $C_\star$ has $K_0$ communities with connectivity $B_\star$. Two regimes are considered: *dense*, with $B_\star$ fixed, and *logarithmic degree*, with $B_\star = (\log n/n)\, Q$. The central structural notion is the following.

**Definition 2.1** (invisible split)**.** Two communities $H, R$ of $C_\star$ form an *invisible split* if $|H| = |R|$, $B_{\star,HH} = B_{\star,RR}$ and $B_{\star,Hl} = B_{\star,Rl}$ for all $l \notin \{H, R\}$. In words, $H$ and $R$ have the same size and internal density, and connect to the rest of the network in the same way.

In the planted partition model ($B_\star = q\mathbf{1}\mathbf{1}^\top + (p-q)I$) every pair of communities forms an invisible split. Hierarchical community structure (D4, F9) generates them as sub-communities.

**Definition 2.2** (merged-pair coarsening and separation time)**.** $C_{HR}$ is the partition obtained from $C_\star$ by merging $H$ and $R$ into one block $M$. The *separation time* $\tau_{\text{sep}}$ is the first time the chain has $K_0$ blocks of size at least $n/(2K_0)$ with distinct majority communities.

## 2.3 Heuristic analysis

#### The price of one new edge probability

For $N$ trials with $s = \hat{\theta}N$ successes and $\hat{\theta} \in (0,1)$ fixed, Stirling's formula gives

$$\log \frac{\mathrm{B}(a+s, b+N-s)}{\mathrm{B}(a,b)} = s\log\hat{\theta} + (N-s)\log(1-\hat{\theta}) - \tfrac{1}{2}\log\frac{N}{2\pi\hat{\theta}(1-\hat{\theta})} + \log\pi_{a,b}(\hat{\theta}) + o(1), \tag{2}$$

where $\pi_{a,b}$ is the Beta$(a,b)$ density. Explaining the same trials by a rate $\bar{\theta}$ that is already well estimated, the log Bayes factor for a *new* parameter is therefore

$$N\,\mathrm{KL}(\hat{\theta}\,\|\,\bar{\theta}) \;-\; c(N,\hat{\theta}), \qquad c(N,\theta) = \tfrac{1}{2}\log\frac{N}{2\pi\theta(1-\theta)} - \log\pi_{a,b}(\theta).$$

If the new parameter has no signal, $N\,\mathrm{KL}(\hat{\theta}\|\bar{\theta})$ is approximately $\frac{1}{2}\chi^2_1$, with mean $\frac{1}{2}$.

- *Dense regime.* The net price is $\frac{1}{2}\log N + O(1)$.
- *Logarithmic degree.* With $\theta \asymp \log n/n$, $N = n/K$ and the uniform prior, the net price is $\log n - \frac{1}{2}\log(K\log n) + O(1)$, about twice as large. For a Beta$(a,b)$ prior the leading term is $a\log n$, so here the price depends on how the prior behaves near zero.

Appendix A.1 gives the details and an exact identity for the uniform prior, which is used for small counts.

#### The opening log-odds

Let $i \in H$ sit in the merged block $M = H \cup R$ of $C_{HR}$. The other blocks are the true communities $l \notin \{H, R\}$. Moving $i$ into a new block creates one new parameter for each existing block $l$, with $n_l$ trials. It removes nothing, because $i$'s edges were explained by the well-estimated rates of $M$. Hence

$$D_i \;=\; \sum_l \Big[ n_l \,\mathrm{KL}(\hat\theta_{il} \,\|\, \bar\theta_{Ml}) - c(n_l, \hat\theta_{il}) \Big] \;+\; \log \frac{w_{\text{new}}}{w_M} \;+\; o(1). \tag{3}$$

- *Invisible split.* The rate of $i$ into every block equals the rate of $M$ into that block. For $l = M$, the rate of $i$ into $M$ is $(p+q)/2$, which is also $M$'s internal average. So every KL term is pure noise, with mean $\frac{1}{2}$.
- *Dense planted partition.* This gives

$$\mathbb{E}D_i \approx -\frac{K_0 - 1}{2} \log \frac{n}{K_0} \;-\; \log \frac{2n}{K_0 \alpha} \;+\; O(K_0), \tag{4}$$

  that is, a factor of about $n^{-(K_0+1)/2}$, with constants exponential in $K_0$. Appendix A.2 gives the constant. With MFM weights the prior term is $-(1+\gamma)\log n + O(1)$ instead, so the factor is $n^{-(K_0+1)/2-\gamma}$; the rest of this section is written for the DP.
- *Average versus maximum.* What drives the sampler is the average opening odds $\lambda = \frac{1}{n_H}\sum_{i\in H} e^{D_i}$. Because the noise in $D_i$ is heavy-tailed, $\lambda = e^{\mathbb{E}D}(\log n)^{O(K_0)}$, while the best-placed nodes exceed $\mathbb{E}D$ by about $\log n_H$ (Appendix A.3).

#### The nucleus

After a node opens a new block $\mathcal{N}$, the block only pays off once it holds $m$ nodes of $H$, so that the $\binom{m}{2}$ pairs inside $\mathcal{N}$ reveal the within-community rate $p$ against $M$'s average $\bar p$. Let $S_m$ be the set of partitions obtained from $C_{HR}$ by moving $m$ nodes of $H$, or $m$ nodes of $R$, into a new block, and let $F(m) = -\log\{\Pi_n(S_m)/\Pi_n(C_{HR})\}$ be its free energy, so that $F(1) = -\log(2n_H\lambda)$. Counting the likelihood, the DP prior and the $2\binom{n_H}{m}$ ways to choose the nucleus (Appendix A.4), for $1 \le m \ll n^{1/3}$,

$$F(m) - F(1) \;\approx\; (m-1)\log\frac{n_M}{n_H} \;-\; \frac{m(m-1)}{2}\,\mathrm{KL}(p \,\|\, \bar p) \;+\; O(K_0 \log m).$$

The first term is the prior's preference for the large block, net of entropy. The second is the signal inside the nucleus. $F$ is maximal at some $m^* \ge \log(n_M/n_H)/\mathrm{KL}(p\|\bar p)$ that does not depend on $n$.

- *Merged pair*, $n_M/n_H = 2$: the nucleus adds an amount that does not depend on $n$, about $(\log 2)^2/(2\mathrm{KL}) + O(K_0 \log m^*)$. The $n$-dependence of the barrier is that of the opening cost.
- *Dilution*: one block holds all $K_0$ communities, so $n_M/n_H = K_0$. The nucleus then adds about $(\log K_0)^2/(2\mathrm{KL})$, which matters for the underestimation law.

#### From costs to times

An update picks a node of $H$ or $R$ with probability $2/K_0$, and the move opens a block with probability about $e^{D_i}$. The new block then grows to $m^*$ with some probability $\rho$ that, according to the heuristic, does not depend on $n$. A birth–death approximation of the nucleus size (Appendix A.5) gives

$$\mathbb{E}\,\tau_{\text{sep}} \;\approx\; \frac{K_0}{2\lambda\rho} \;=\; n^{(K_0+1)/2}\,(\log n)^{O(K_0)}\,e^{O(K_0 \log K_0)} \text{ updates} \;=\; n^{(K_0-1)/2}\,(\log n)^{O(K_0)}\,e^{O(K_0 \log K_0)} \text{ sweeps.} \tag{5}$$

This is the *exponent law*. The heuristic predicts the exponent of $n$, not the constants. In the simulated range the measured $\tau\lambda$ still decreases with $n$, which is consistent with $\rho$ increasing and lowers the apparent exponent (Section 2.4). Under MFM weights the exponent is $(K_0+1)/2 + \gamma$. In the logarithmic-degree regime the price per parameter doubles for the uniform prior, and the exponent should approach $K_0$ in updates; for a Beta$(a, b)$ prior it should be $a(K_0 - 1) + 1$ (Appendix A.1).

#### The underestimation law

From the one-block partition the chain peels off one community at a time, and the $j$-th opening happens inside a block while $j$ blocks exist. It therefore creates $j$ new parameters and costs about $\frac{j+2}{2}\log n$, up to terms that depend only on $K_0$ and $j$, plus a dilution term; it takes about $n^{(j+2)/2}$ updates. With a budget of $T = n^\beta$ updates, the number of communities found is at most about $2(\beta - 1) + O(1)$, independently of $K_0$ (Appendix A.6). Under MFM weights the bound becomes $2(\beta - \gamma) - 1 + o(1)$.

#### Visible splits

If $B_{\star,Hl} \neq B_{\star,Rl}$ for some $l$, the KL terms in (3) have means of order $(n/K_0)\,\delta_l^2/\theta_l$, where $\delta_l = |B_{\star,Hl} - B_{\star,Rl}|$. Each node of $H$ then carries a signal, and the nucleus grows without a barrier once $\sum_l (n/K_0)\delta_l^2/\theta_l \gtrsim K_0 \log n$. Once a block has opened, a much weaker signal already makes the nucleus grow (Appendix A.8). The crossover between the two regimes is a natural target (C1.4).

## 2.4 Numerical evidence

All runs use Beta(1,1) and DP(1). Unless stated otherwise the model is the planted partition with $p = 0.5$ and $q = 0.05$. Code and raw outputs are described in Appendix B.

| Setting | $n$ | $K_0$ | predicted $\mathbb{E}D$ | measured mean (max) | (S1) separates merged pair |
|---|---|---|---|---|---|
| fixed $n$ | 800 | 2 | $-9.4$ | $-9.5$ ($-5.5$) | 1/2 (30 sweeps) |
| | 800 | 4 | $-13.9$ | $-14.2$ ($-9.4$) | 0/2 |
| | 800 | 8 | $-21.9$ | $-22.0$ ($-15.5$) | 0/2 |
| | 800 | 16 | $-36.0$ | $-36.1$ ($-29.9$) | 0/2 |
| fixed block size 100 | 200 | 2 | $-7.3$ | $-7.5$ ($-5.4$) | 1/3 (50 sweeps) |
| | 400 | 4 | $-12.2$ | $-12.2$ ($-9.2$) | 0/3 |
| | 800 | 8 | $-21.9$ | $-21.9$ ($-15.5$) | 0/3 |
| | 1600 | 16 | $-41.4$ | $-41.9$ ($-33.8$) | 0/3 |
| fixed $K_0 = 4$ | 200 | 4 | $-10.5$ | $-10.5$ ($-7.3$) | 0/4 (100 sweeps) |
| | 400 | 4 | $-12.2$ | $-12.2$ ($-9.2$) | 0/4 |
| | 800 | 4 | $-13.9$ | $-14.0$ ($-10.4$) | 0/4 |
| | 1600 | 4 | $-15.6$ | $-15.8$ ($-10.2$) | 0/4 |
| log degree | 800 | 2 | $-10.2$ | $-10.1$ ($-6.6$) | 1/2 (50 sweeps) |
| $((\sqrt{\tilde{a}} - \sqrt{\tilde{b}})^2 = 2.25K_0)$ | 800 | 4 | $-16.3$ | $-16.6$ ($-10.4$) | 0/2 |
| | 800 | 8 | $-28.1$ | $-27.8$ ($-21.8$) | 0/2 |

**Table 2:** Opening log-odds. The prediction is formula (3) evaluated exactly, by summing over binomial counts, with no fitted constants. The measured values are exact, from the merged-pair coarsening: mean and maximum over every second node of one merged community (all its nodes in the first block of rows). The 15 rows contain 12 distinct settings. In every run started from the truth, (S1) stayed at the truth.

#### Reading the evidence.

- *The cost formula is quantitatively right* (Table 2). The opening log-odds match the prediction to within about 0.5 nats in all 15 rows. At fixed block size they grow by 2.46 nats per extra community, against 2.43 from the exact predictions (2.41 asymptotically). At $K_0 = 4$ they grow by 2.55 per unit of $\log n$, against 2.45 exact (2.5 asymptotically).
- *The opening rate has the predicted exponent; the hitting times grow more slowly at these sizes* (Table 3).
  - The average opening odds scale as $n^{-1.52}$ and $n^{-1.94}$ for $K_0 = 2$ and 3, against $n^{-1.50}$ and $n^{-1.98}$ predicted.
  - The mean hitting times scale as $n^{1.39}$ and $n^{1.56}$ updates, below the predicted exponents. For $K_0 = 3$ the 95% interval excludes the prediction.
  - The difference lies in the nucleus factor. $\tau\lambda$, which estimates $K_0/(2\rho)$, still decreases with $n$, which is consistent with a newly opened block surviving more often in larger graphs. Candidate causes are the finite-size terms of order $m^{*2}/n_H$ and $m^{*3}/n_H$ in the nucleus free energy (with $n_H$ between 50 and 800 here; the two act in opposite directions) and graph-to-graph variation in $\rho$. Hits within six sweeps are also more frequent than an exponential law would give (3 of 32 runs at $K_0 = 3$, $n = 300$ and 600, against about 0.2 expected); their graphs have only slightly larger $\lambda$. This is to be resolved by measuring $\rho(n)$ directly.
  - The decrease must stop eventually. Separation needs at least one opening, which takes about $K_0/(2\lambda)$ updates, so heuristically $\tau\lambda$ stays above about $K_0/2$, and the asymptotic exponent is at least $(K_0+1)/2$. Extrapolating the trend for $K_0 = 3$, this floor is reached only near $n \approx 6 \times 10^4$.
  - For $K_0 \geq 4$ the predicted times exceed the budgets, and indeed no run separated the pair.

| | | | | | |
|---|---|---|---|---|---|
| $K_0 = 2$, *from one block* (E3, E3b) | | | | | |
| $n$ | 100 | 200 | 400 | 800 | 1600 |
| runs | 40 | 40 | 40 | 40 | 24 |
| hitting time in sweeps, median (mean) | 25.5 (34.1) | 25 (40.4) | 33 (47.0) | 53.5 (98.6) | 64 (83.8) |
| log of the mean of $\tau\lambda$ | 2.0 | 1.9 | 1.6 | 2.0 | 1.5 |
| $K_0 = 3$, *from the merged pair* (E7) | | | | | |
| $n$ | 150 | 300 | 600 | 1200 | |
| runs | 16 | 16 | 16 | 12 | |
| hitting time in sweeps, median (mean) | 136 (282) | 523 (815) | 516 (841) | 760 (1025) | |
| log of the mean of $\tau\lambda$ | 3.0 | 3.1 | 2.6 | 2.2 | |

| Exponent of $n$, in updates | $K_0 = 2$ | $K_0 = 3$ |
|---|---|---|
| predicted for the opening rate and the hitting time | 1.50 | 1.98 |
| measured: average opening odds, $-\log\lambda$ | 1.52 | 1.94 |
| measured: mean hitting time, with 95% bootstrap interval | 1.39 (1.22–1.54) | 1.56 (1.22–1.93) |
| measured: $\tau\lambda$, the nucleus factor | $-0.13$ | $-0.45$ |

**Table 3:** Hitting times of the truth under (S1). Every run reached the truth within its budget (300 or 1000 sweeps for $K_0 = 2$, 4000 for $K_0 = 3$). $\tau$ is the hitting time in updates and $\lambda$ the exact average opening odds of the same graph at $C_{HR}$. The predicted exponents use the exact $\mathbb{E}D$ of (3); asymptotically they are $(K_0+1)/2 = 1.5$ and 2.

| | $K_0 = 4$ | $K_0 = 8$ | $K_0 = 16$ | $K_0 = 16$, long run |
|---|---|---|---|---|
| budget | 100 sweeps | 100 sweeps | 100 sweeps | 1000 sweeps ($8 \times 10^5$ updates) |
| communities found (two runs) | 2 and 3 | 1 and 2 | 1 and 2 | 2 and 2 |

**Table 4:** Underestimation from the default one-block start ($n = 800$). The long runs recorded 2 communities at 10, 30, 100, 300 and 1000 sweeps (run 1), and 1, 1, 1, 2, 2 (run 2).

- *Underestimation is severe and does not depend on* $K_0$ (Table 4). It is consistent with the underestimation law, which allows at most about 2 openings for $T = 8 \times 10^5 \approx n^{2.03}$.
- *The obstruction is specific* (Tables 5 and 6). Split–merge removes it, and visible splits do not have it.

## 2.5 Precise targets

Throughout, "with high probability" refers to the random graph, and the samplers are (S1) and (S2) of Section 2.2. The targets in C1 are stated for the DP prior. Under an MFM prior with parameter $\gamma$ the new-block weight carries an extra factor $n^{-\gamma}$ (Section 2.2), so every exponent in C1 increases by $\gamma$, and the bound in Target 2.3 becomes $2(\beta - \gamma) - 1$. Constants may depend on $a, b, \alpha$ (or $\gamma$ and $p_K$) and on $B_\star$, but not on $n$. These are *targets*: statements to be proved, sharpened or refuted.

**Assumption 2.1** (planted partition)**.** $C_\star$ has $K_0$ communities of equal size $n/K_0$ and connectivity $B_\star = q\mathbf{1}\mathbf{1}^\top + (p-q)I$, with $p \neq q$. Either (dense) $p, q \in (0,1)$ are fixed, or (logarithmic degree) $p = \tilde{a}\log n/n$ and $q = \tilde{b}\log n/n$ with $(\sqrt{\tilde{a}} - \sqrt{\tilde{b}})^2 > K_0(1+\epsilon)$, which is above exact recovery (T4). (The tildes distinguish these constants from the Beta hyperparameters $a, b$.)

#### C1: single-site samplers cannot open communities

**Target 2.1** (C1.1: exponent law, $K_0$ fixed)**.** Under Assumption 2.1 (dense), start (S1) at a merged-pair coarsening $C_{HR}$. For every $\epsilon > 0$, with high probability,

$$n^{\frac{K_0+1}{2}-\epsilon} \;\le\; \mathbb{E}\,\tau_{\text{sep}} \;\le\; n^{\frac{K_0+1}{2}+\epsilon} \quad \text{updates.}$$

In the logarithmic-degree regime the exponent $(K_0+1)/2$ is replaced by $\theta_{\log}(K_0)$, conjecturally $a(K_0-1)+1$ for Beta$(a,b)$ priors, that is $K_0$ for the uniform prior (Appendix A.1).

**Target 2.2** (C1.2: superpolynomial barrier for growing $K_0$)**.** If $K_0 = K_n \to \infty$ with $K_n \le n^\delta$ for a small $\delta > 0$ (dense regime), then with high probability

$$\mathbb{P}_{C_{HR}}\big(\tau_{\text{sep}} \le e^{(1-\epsilon)\frac{K_n}{2}\log\frac{n}{K_n}}\big) \to 0.$$

| (S2) (3 spectral split–merge proposals per sweep), $n = 800$ | $K_0 = 4$ | $K_0 = 8$ | $K_0 = 16$ |
|---|---|---|---|
| from the merged pair: sweep at which the truth is reached | 1, 1 | 1, 15 | 1, 16 |
| from one block: sweep at which the truth is reached | 5, 1 | 16, 10 | not in 30 (15 and 14 of 16 found) |

**Table 5:** Split–merge recovers what single-site moves cannot. Each cell gives two runs.

| Split (random heterogeneous $B_\star$) | opening log-odds mean (max) | (S1) from merged pair | (S2) |
|---|---|---|---|
| visible (generic $B_\star$) | $-11.7$ ($-1.8$) | all 8 found at sweep 1 (2/2) | — |
| invisible (rows of $H, R$ made equal) | $-19.7$ ($-14.6$) | stuck for 30 sweeps (2/2) | sweeps 1 and 8 |

**Table 6:** The dichotomy ($K_0 = 8$, $n = 800$). The other six communities and $H$ keep the same random connectivity. Setting $R$'s row equal to $H$'s makes the two indistinguishable from outside, and turns an immediate separation into a stuck chain. The visible runs found all eight communities with one extra small block. The invisible case also sets $B_{\star,HH} = B_{\star,RR} = 0.5$ (required by Definition 2.1) and lowers $B_{\star,HR}$ from 0.25 to 0.1. The latter strengthens the signal between $H$ and $R$, so it cannot explain the stuck chain. The log-posterior gap between the truth and the merged state is 1875 nats in the invisible case.

For example, $K_n \asymp \log n$ gives $n^{c \log n}$, and $K_n = n^\delta$ gives $e^{cn^\delta \log n}$.

**Target 2.3** (C1.3: underestimation law)**.** Start (S1) at the one-block partition and fix $\beta > 1$. With high probability, after $T = n^\beta$ updates the number of blocks of size at least $n/(2K_0)$ is at most $k(\beta)$. Here $k(\beta) < \infty$ does not depend on $K_0$, and heuristically $k(\beta) \le 2(\beta - 1) + O(1)$ (Section 2.3). In particular, if $K_0 \to \infty$ the fraction of communities discovered in polynomial time tends to zero.

**Target 2.4** (C1.4: visible versus invisible splits)**.** Let $H, R$ be two communities of a general $B_\star$, with $\delta_l = |B_{\star,Hl} - B_{\star,Rl}|$.

(a) If $(H, R)$ is an invisible split (Definition 2.1), Target 2.1 holds with exponent $(K_0 + 1)/2$.

(b) If $\sum_l \frac{n}{K_0} \delta_l^2 / (B_{\star,Hl}(1 - B_{\star,Hl})) \ge CK_0 \log n$, then (S1) separates $H$ and $R$ from $C_{HR}$ in $O(n \log n)$ updates with high probability.

(c) Between the two, the exponent decreases continuously with the visible signal.

**Target 2.5** (C1.5: split–merge is polynomial)**.** Under Assumption 2.1, (S2) started from any coarsening of $C_\star$ (including one block) hits $C_\star$ within $\mathrm{poly}(n)$ updates with high probability. Started from a merged-pair coarsening, $O(K_0)$ split–merge proposals in expectation plus $O(\log n)$ sweeps suffice. For a lazy, reversible version of (S2) (a random mixture of the split–merge and single-site kernels), the hitting-to-mixing reduction (G4) together with Target 2.6 would make the mixing time polynomial, provided the hitting bound for large sets holds from *every* starting partition, not only from coarsenings of $C_\star$.

#### C2: what the posterior sees

**Target 2.6** (C2.1: posterior concentration with growing $K_0$)**.** Under Assumption 2.1 with $K_0 \le n^\delta$, DP or MFM priors, and standard Beta priors, $\Pi_n(C = C_\star) \to 1$ in probability.

This extends B14, which covers $K_0 \in \{2, 3\}$, and B17, which needs an assortative prior. The growth rate must respect the statistical limits for many communities (T2, T8, D3).

**Target 2.7** (C2.2: Dirichlet-process SBMs are consistent for $K$)**.** Under Assumption 2.1 and a DP($\alpha$) prior with $\alpha$ fixed, $\Pi_n(K = K_0) \to 1$.

*Reason* (Appendix A.7). Move $m$ nodes of one community of $C_\star$ into a new block. The DP prior times the number of ways to choose them is about $\alpha/m$, so the prior alone does not penalise an extra block. The likelihood does: the new block needs $K_0$ new edge probabilities, each estimated from at least $n/K_0$ trials, without signal. Summed over sizes and communities, the posterior odds of one extra block are $O(K_0 \alpha (n/K_0)^{-K_0/2} (\log n)^{O(K_0)}) \to 0$. In fixed-dimensional mixtures a one-point cluster costs only $O(1)$ in likelihood, so the extra-cluster odds stay of order $\alpha$, which is why DP mixtures are inconsistent (S8). Numerically, the total odds of an extra one-node block at the truth, summed over all nodes, are $e^{-3.2}$, $e^{-5.8}$, $e^{-13.0}$ and $e^{-24.7}$ for $K_0 = 2, 4, 8, 16$ ($n = 800$).

#### C3: local mixing with random $K$

**Target 2.8** (C3: rapid mixing near the truth)**.** Under Assumption 2.1, start (S1) at any $C_0$ whose blocks match $C_\star$ up to $\gamma_0 n$ misclassified nodes, with $\gamma_0 = o(1)$, plus at most $k$ spurious blocks of total size $o(n)$. Then $\tau_\epsilon(C_0) \lesssim n^2(n^2 I + \log \epsilon^{-1})$, as in Zhuo & Gao (M1), without restricting to balanced partitions. Spurious blocks are emptied within $O(n \log n)$ updates.

#### C4: the opening-cost principle

**Target 2.9** (C4: general lower bound)**.** For collapsed conjugate exponential-family partition posteriors, the log-odds that a single-site move opens a block from one unit are at most minus the information cost of the new parameters given that unit, plus the unit's own information about them. The intended corollaries are:

- $e^{c(\tau)p}$ for isotropic Gaussian mixtures (P3.A);
- $n^{(K_0+1)/2}$ for SBMs (Target 2.1);
- $n^{cKL}$ for multilayer SBMs (P3.B);
- $n^{cK_c}$ to open a row cluster and $n^{cK_r}$ to open a column cluster in latent block models (B12).

## 2.6 Proof strategy

#### Building blocks

**Lemma to prove 2.1** (uniform Beta asymptotics)**.** Formula (2) holds with explicit error $O(1/\min(s, N-s))$, uniformly over $1 \le s \le N-1$. For small counts the exact form is $\log\{s!(N-s)!/(N+1)!\}$ when $a = b = 1$. *Tools:* Stirling bounds with remainders.

**Lemma to prove 2.2** (edge counts are uniformly concentrated)**.** With high probability, simultaneously for all disjoint node sets $S, T$ of size at most $m$, $E(S,T)$ is within $O(\sqrt{|S||T|\bar{\theta}\, m \log n} + m \log n)$ of its mean, where $\bar{\theta}$ is the average connection probability between $S$ and $T$. *Tools:* Bernstein's inequality and a union bound with $\log \binom{n}{m} \le m \log(en/m)$.

**Lemma to prove 2.3** (the opening bound)**.** Under an invisible split, with high probability the average opening odds satisfy

$$\log\Big(\frac{K_0}{n}\sum_{i\in H} e^{D_i}\Big) \le -\frac{K_0-1}{2}\log\frac{n}{K_0} - \log\frac{2n}{K_0\alpha} + O(K_0 \log\log n) + O(\sqrt{K_0 \log n}),$$

and the same holds for $R$. *Tools:* Lemmas 2.1–2.2 applied to (3). The KL terms are approximately independent $\frac{1}{2}\chi^2_1$ variables. Their exponentials are heavy-tailed, so the sum must be controlled by truncation at the level of the largest term, not through its mean (Appendix A.3). A bound on $\max_i D_i$ alone does not suffice: $e^{\max_i D_i}$ exceeds the average by a factor of order $n/K_0$, up to logarithmic factors.

**Lemma to prove 2.4** (nucleus and bottleneck)**.** Let $\mathcal{B}$ be the set of partitions in which at most one block contains $m^*$ or more nodes of $H \cup R$. It contains $C_{HR}$, and every partition counted by $\tau_{\text{sep}}$ lies outside it: that needs two blocks of size at least $n/(2K_0)$ with majorities $H$ and $R$, and each holds at least $n/(2K_0^2) \ge m^*$ nodes of $H \cup R$. Leaving $\mathcal{B}$ requires a second block, pure or mixed, with $m^*$ nodes of $H \cup R$. Then with high probability $Q(\mathcal{B}, \mathcal{B}^c)/\Pi_n(C_{HR}) \le n^{-\frac{K_0+1}{2}+\epsilon}$ (Appendix A.5). *Tools:*

- the free-energy computation of Section 2.3;
- a Peierls-type summation over the rest of the partition, which is dominated by $C_{HR}$;
- templates: the planted-model conductance argument of M7, Jerrum's gateway argument (M3), and the conductance bounds of M14.

Combined with $\mathbb{P}_x(\tau_{\mathcal{B}^c} \le t) \le t\, Q(\mathcal{B}, \mathcal{B}^c)/\Pi_n(x)$, this proves the lower bound of Target 2.1.

**Lemma to prove 2.5** (matching upper bound)**.** From $C_{HR}$, the escape probability per update is at least $n^{-\frac{K_0+1}{2}-\epsilon}$, and once the nucleus exceeds $m^*$, absorption into the separated state takes $O(n \log n)$ updates. *Tools:*

- an explicit path (open, grow, absorb) with a lower bound on its probability;
- a drift argument for growth beyond $m^*$, where the gain is quadratic in $m$;

- the local-majority property of exact recovery (T7) for the final clean-up.

**Lemma to prove 2.6** (spectral splits)**.** Under Assumption 2.1, consider a block that is a union of $r \geq 2$ communities, and split it by the sign of the leading eigenvector of its centred adjacency submatrix. For $r = 2$ the split is exact up to $o(n/K_0)$ nodes. For $r \geq 3$ the leading eigenvector is approximately constant on each community. The split keeps every community whose coefficient is not close to zero on one side, up to an $o(1)$ fraction of its nodes, and may cut a community whose coefficient is close to zero; a later merge or Gibbs sweep repairs it. *Tools:* Davis–Kahan and entrywise eigenvector bounds (G5), and the recursive-partitioning analysis of F9 and F1.

**Lemma to prove 2.7** (posterior ratios)**.** For partitions at distance $d$ from $C_\star$, in the sense of misclassified nodes and extra or missing blocks,

$$\log \frac{\Pi_n(C)}{\Pi_n(C_\star)} \leq -c\, d \, \min\left\{\frac{nI}{K_0}, \log n\right\} \quad \text{w.h.p.}$$

*Tools:* the posterior-ratio lemmas of M1 and B14, and the penalised-evidence analysis of S5. This yields Targets 2.6 and 2.7, and with Lemma 2.3 it gives Target 2.8.

#### Assembling the targets

- *Target 2.1:* Lemmas 2.1, 2.2, 2.3, 2.4 for the lower bound; Lemma 2.5 for the upper bound.
- *Target 2.2:* the same lemmas with $K_0$-uniform constants. The union bounds must survive $K_0 \leq n^\delta$.
- *Target 2.3:* apply Lemma 2.4 at each stage, with the dilution term from Section 2.3.
- *Target 2.4:* Lemma 2.3 with the mean of the KL terms included. Part (b) follows from a positive drift for the nucleus starting at $m = 1$.
- *Target 2.5:* Lemma 2.6 for the split, Lemma 2.7 for acceptance, and Target 2.8 for the clean-up, followed by the reduction of mixing to hitting times (G4).
- *Targets 2.6–2.7:* Lemma 2.7, summed over partitions using Lemma 2.2.

## 2.7 Relation to existing work

| Closest work | What it does | What P3.C adds |
|---|---|---|
| Zhuo & Gao (M1) | rapid mixing near the truth, $K$ known | random $K$ (C3), and the global picture: slow creation (C1) and fast split–merge (C1.5) |
| Riolo et al.; Peixoto; Newman & Reinert (D9, D7, D8) | document failures empirically | a mechanism, exact cost formulas, exponents and an underestimation law |
| Côme & Latouche (B10) | document over-retention of clusters | a separation of the two directions: creation is polynomial of growing degree, merging is close to neutral drift (Section 2.8) |
| Geng et al.; Jiang & Tokdar; Gao et al. (B14, B17, S2) | posterior theory for $K$ in special cases | concentration with growing $K_0$ and standard priors; DP consistency (C2) |
| Potts metastability (M5, M7) | first-order barriers in statistical physics | the first barrier theorem for a Bayesian clustering posterior with unknown $K$ |
| Planted clique (M3, M4) | local chains miss planted structure | the barrier sits in a regime that is statistically *and* computationally easy, so it is purely a sampler artefact |
| Tosh & Dasgupta; P3.A (M14) | barriers for mixtures | a network barrier with a different source: the number of communities, not the dimension |

**Table 7:** Novelty check.

**Why this suits a top journal.**

- It explains a widely observed failure of widely used software with a clean, testable law. The cost formula already matches simulations to within half a nat.
- It separates statistical, computational and *algorithmic* difficulty. The posterior and spectral methods succeed where single-site samplers fail.
- It gives practical guidance: initialisation, split–merge, and what diagnostics can and cannot detect.

Target venues are AoS, JASA (Theory and Methods) and JRSS-B for C1 and C2, and JMLR or Bernoulli for C3.

## 2.8 Risks, fallbacks and the second direction

- **Sharp exponents may be hard.**
  - The nucleus phase combines likelihood, the preferential attachment of the prior, and entropy.
  - *Fallback:* prove $\mathbb{P}_{C_{HR}}(\tau_{\text{sep}} \le n^{cK_0}) \to 0$ for some $c > 0$, with constants uniform in $K_0 \le n^{\delta}$. This gives a weaker form of Target 2.2, with $c$ in place of $(1-\epsilon)/2$, and its stage-by-stage version, with the dilution terms, gives a weaker underestimation law. Then state the exponent $(K_0+1)/2$ as a conjecture.
  - The lower half is the safer one. Heuristically it is governed by the opening rate, whose exponent Table 3 confirms, but a proof still needs the summation over $\mathcal{B}$ in Lemma 2.4.
- **Pre-asymptotic exponents.** At $n \le 1600$ the measured hitting-time exponents (1.39 and 1.56 for $K_0 = 2, 3$) are below $(K_0+1)/2$. The measured $\tau\lambda$ still decreases with $n$, which is consistent with the survival probability $\rho$ of a new block growing with $n$ (Table 3), but the cause has not been isolated. The theorem would describe the asymptotic regime, while practitioners see the pre-asymptotic one. Understanding $\rho(n)$, which is part of Lemma 2.5, is therefore part of the project, and the paper should report both.
- **Small $K_0$ gives moderate barriers.** For $K_0 = 2$, single-site Gibbs separated the pair within 455 sweeps in all 184 hitting-time runs up to $n = 1600$ (E3, E3b). The striking statements are those for growing $K_0$ (C1.2, C1.3). The paper should lead with those.
- **Degree correction.** Riolo et al. and Peixoto use degree-corrected SBMs, where each node has its own degree parameter. Opening a block then also re-estimates degree propensities. The cost formula should carry over with modified constants. This needs to be checked before the paper claims to explain those papers' observations.
- **Prior choices.**
  - MFM weights change $\log(w_{\text{new}}/w_M)$ by $-\gamma \log n + O(1)$, where only the $O(1)$ term depends on $p_K$. The exponents then increase by $\gamma$. Geng et al. (B14) use this prior, so the paper must treat it explicitly.
  - Gibbs-type priors (B18) change how the opening weight depends on the number of blocks.
  - Beta hyperparameters change constants only.

  The results should be stated for a class of priors.
- **The second direction: too many communities** (B10).
  - Merging two spurious blocks of the same community by single-site moves is nearly neutral: the likelihood is flat between the blocks, and only the prior and the final parameter savings push towards merging. It behaves roughly like a Moran model, so it needs about $n_{\text{block}}^2$ updates of the blocks' own nodes, about $n\, n_{\text{block}}$ single-site updates in total. That is polynomial, although the drift is not exactly neutral.
  - Many *small* spurious blocks in sparse graphs can be slow to absorb. This is consistent with Côme & Latouche's report, and it is a separate, secondary target.
- **Competition.** None was found on this question. The nearest active lines are Glauber dynamics for two-community inference (M12) and non-reversible clustering samplers (X4), neither of which addresses the creation of communities.

## 2.9 Work plan and paper outline

1. **Months 1–3.** Lemmas 2.1–2.3 and the lower half of Target 2.1 for $K_0$ fixed (dense). Extend the simulations: larger $n$ for $K_0 = 3$ and 4, a direct measurement of the survival probability $\rho(n)$ of a new block, the logarithmic-degree regime, MFM weights, and degree correction.

2. **Months 3–6.** Lemma 2.4 with $K_0$-uniform constants, giving Targets 2.2 and 2.3. Lemma 2.6 and Target 2.5.
3. **Months 6–9.** Target 2.8 (C3) and Targets 2.6–2.7 (C2). Write-up.
4. **Later.** The matching upper bound (Lemma 2.5), visible splits (Target 2.4), and the umbrella principle (C4) together with P3.A and P3.B.

**Suggested outline of the main paper.**

1. Introduction: the empirical puzzle (D9, D7), with Table 4 as a teaser.
2. Model, samplers and invisible splits.
3. The opening cost: exact formula and its asymptotics (Table 2).
4. Lower bounds: the exponent law and superpolynomial barriers.
5. The underestimation law.
6. Upper bounds: split–merge and local mixing.
7. Posterior concentration and DP consistency.
8. Simulations and practical recommendations.
9. Discussion: degree correction, hierarchies and other models.

## 2.10 Extensions and open questions

- **Hierarchical (nested) SBMs** (D4, F9). Every sub-community split is invisible by construction. Do nested priors make it easier or harder to open such splits?
- **Below exact recovery.** The posterior no longer concentrates on $C_\star$. What does a single-site sampler explore between Kesten–Stigum and exact recovery (T15, T16)?
- **Multilayer SBMs** (P3.B). The opening cost becomes $K_0 L$ parameters, so barriers of order $n^{cK_0L}$ are expected.
- **Latent block models and bipartite networks** (B12). Opening a row cluster costs $K_c$ parameters, and a column cluster $K_r$.
- **Informed and non-reversible proposals.** Locally balanced or multiple-try proposals (M12) and lifted samplers (X4) cannot remove a barrier caused by the posterior itself. Proving this is a clean side result.
- **Diagnostics.** The chain stays at the coarsening $C_{HR}$, a state of high local stability, so traceplots and $\hat{R}$ from similar starts cannot detect it. Coupled chains from over-clustered starts can (X6).

# A Derivations

This appendix gives the computations behind Section 2.3. Everything here is heuristic; the statements that must be proved are listed in Section 2.6. Notation:

- $h = n/K_0$ is the community size, and $n_M = 2h$ is the size of the merged block $M = H \cup R$;
- $\bar{p}$ is the internal edge density of $M$, and $\kappa = \mathrm{KL}(p \,\|\, \bar{p})$ is the Bernoulli Kullback–Leibler divergence;
- BF denotes the Bayes factor of one new edge probability against a known rate (A.1).

## A.1 The price of one new edge probability

**Stirling.** For fixed $c$ and $x \to \infty$, $\log\Gamma(x+c) = (x+c-\frac{1}{2})\log x - x + \frac{1}{2}\log(2\pi) + O(1/x)$. Apply this to the three Gamma functions in $\mathrm{B}(a+s, b+N-s) = \Gamma(a+s)\Gamma(b+N-s)/\Gamma(a+b+N)$, with $s = \hat{\theta}N$. The terms linear in $s$ and $N$ cancel, and

$$\begin{aligned}\log \mathrm{B}(a+s, b+N-s) \;&= s\log\hat{\theta} + (N-s)\log(1-\hat{\theta}) + (a-\tfrac{1}{2})\log\hat{\theta} + (b-\tfrac{1}{2})\log(1-\hat{\theta}) \\ &\quad - \tfrac{1}{2}\log N + \tfrac{1}{2}\log(2\pi) + O\Big(\frac{1}{\min(s, N-s)}\Big).\end{aligned}$$

Subtracting $\log \mathrm{B}(a,b)$ and using $\log \pi_{a,b}(\theta) = (a-1)\log\theta + (b-1)\log(1-\theta) - \log \mathrm{B}(a,b)$ gives (2).

**An exact identity for the uniform prior.** For $a = b = 1$, $\mathrm{B}(1+s, 1+N-s) = s!\,(N-s)!/(N+1)!$. Hence, for any known rate $\bar{\theta}$,

$$\log \mathrm{BF}(s) \;:=\; \log \frac{\mathrm{B}(1+s, 1+N-s)}{\bar{\theta}^s (1-\bar{\theta})^{N-s}} \;=\; -\log(N+1) - \log b_{N,\bar{\theta}}(s), \tag{6}$$

where $b_{N,\bar{\theta}}$ is the Binomial$(N, \bar{\theta})$ probability mass function. Two consequences:

- $\mathbb{E}_{\bar{\theta}}\mathrm{BF} = 1$. This holds for every prior, because the marginal likelihood is a probability distribution. It matters in A.3.
- $\mathbb{E}_{\bar{\theta}} \log \mathrm{BF} = -\log(N+1) + \mathcal{H}(\mathrm{Binomial}(N, \bar{\theta}))$, where $\mathcal{H}$ is the Shannon entropy. This is the exact *null price* of a new parameter, the quantity that the predictions of Table 2 add up.

**Dense regime.** If $\bar{\theta}$ is fixed, $\mathcal{H}(\mathrm{Binomial}(N, \bar{\theta})) = \frac{1}{2}\log(2\pi e N\bar{\theta}(1-\bar{\theta})) + O(1/N)$. The null price is therefore

$$-\mathbb{E}\log \mathrm{BF} \;=\; \tfrac{1}{2}\log \frac{N}{2\pi e\,\bar{\theta}(1-\bar{\theta})} + O(1/N),$$

which equals $c(N, \bar{\theta}) - \frac{1}{2}$ of Section 2.3. For a general Beta$(a,b)$ prior, add $-\log \pi_{a,b}(\bar{\theta})$. For $N = 100$ and $\bar{\theta} = 0.05$ the formula gives 2.41 nats, and the exact value is 2.43.

**Logarithmic degree.** If $\bar{\theta} = \mu/N$ with an expected count $\mu = O(\log n)$, the entropy tends to that of a Poisson$(\mu)$ law, which is about $\frac{1}{2}\log(2\pi e\mu)$ for large $\mu$. For a Beta$(a,b)$ prior, expanding $\Gamma(a+s)$ instead gives

$$-\mathbb{E}\log \mathrm{BF} \;=\; a\log N - (a-\tfrac{1}{2})\log\mu - \tfrac{1}{2}\log(2\pi e) + \log \mathrm{B}(a,b) + o(1).$$

A numerical check with $b = 1$, $N \in \{10^3, 10^4, 10^5\}$ and $\mu \in \{3, 10, 30\}$ confirms the constant: about $-0.73$, $-1.41$, and $-2.1$ to $-2.3$ for $a = \frac{1}{2}$, 1 and 2. Two consequences:

- With $N = n/K$, $\mu \asymp \log n/K$ and the uniform prior, the price is $\log n - \frac{1}{2}\log(K\log n) + O(1)$, as stated in Section 2.3.
- In this regime the leading term of the price depends on how the prior behaves near zero, through $a$ only. The exponent of Target 2.1 should then be $a(K_0 - 1) + 1$ in updates: $K_0$ for the uniform prior, but $(K_0+1)/2$, as in the dense regime, for the Jeffreys prior Beta$(\frac{1}{2}, \frac{1}{2})$.

**Small counts.** When $\mu$ is of order one, the asymptotic forms are inaccurate. An example is the between-community pairs of experiment E4, where $\mu = nq/K_0$ is 1.7 for $K_0 = 4$ and 0.8 for $K_0 = 8$. The predictions then use (6) directly.

## A.2 The opening log-odds

**Exact expression.** Let $i \in H$ sit in $M$ in the coarsening $C_{HR}$, and remove $i$. Write $n_l$ for the block sizes after the removal ($n_M - 1$ for $M$), and $d_{il}$ for the number of edges from $i$ to block $l$. The weights of (S1) give

$$D_i = \log \frac{w_{\text{new}}}{w_M} + \sum_l \left[ \log \frac{\mathrm{B}(a + d_{il}, b + n_l - d_{il})}{\mathrm{B}(a,b)} - \log \frac{\mathrm{B}(a + E_{Ml} + d_{il},\, b + N_{Ml} + n_l - E_{Ml} - d_{il})}{\mathrm{B}(a + E_{Ml},\, b + N_{Ml} - E_{Ml})} \right].$$

The second ratio is the posterior predictive probability of $i$'s $n_l$ new trials, given the $N_{Ml} \approx n_M n_l$ trials that the pair $(M, l)$ already has.

- A Laplace expansion shows that it equals the plug-in likelihood at $\hat\theta_{Ml} = E_{Ml}/N_{Ml}$, up to a factor $1 + O(n_l/N_{Ml}) = 1 + O(1/n_M)$.
- $\hat\theta_{Ml}$ is within $O_P(N_{Ml}^{-1/2})$ of the true rate $\bar\theta_{Ml}$. Replacing one by the other changes the log-likelihood of $n_l$ trials by $O_P((n_l/N_{Ml})^{1/2}) = O_P(n_M^{-1/2})$.

Hence $D_i = \log(w_{\text{new}}/w_M) + \sum_l \log \mathrm{BF}_l(d_{il}) + o_P(1)$, where $\mathrm{BF}_l$ is the Bayes factor of A.1 for $n_l$ trials against the known rate $\bar\theta_{Ml}$. This is (3).

**Planted partition.** Two kinds of terms appear.

- For each of the $K_0 - 2$ true communities $l \notin \{H, R\}$: $d_{il} \sim \text{Binomial}(h, q)$, and $\bar\theta_{Ml} = q$.
- For $l = M$: $d_{iM}$ is the sum of independent $\text{Binomial}(h-1, p)$ and $\text{Binomial}(h, q)$ counts. Its mean rate per trial, $((h-1)p + hq)/(2h-1)$, equals $M$'s internal density $\bar p = [h(h-1)p + h^2 q]/[h(2h-1)]$ *exactly*. So there is no signal, only a smaller variance.

The predictions in Table 2 replace the law of $d_{iM}$ by $\text{Binomial}(2h-1, \bar p)$ and evaluate every $\mathbb{E}\log \mathrm{BF}_l$ exactly with (6). With the DP weights $w_{\text{new}}/w_M = \alpha/(2h-1)$:

$$\mathbb{E}D \;=\; \log \frac{\alpha}{2h-1} - (K_0 - 2)\Big[\log(h+1) - \mathcal{H}\big(\text{Bin}(h,q)\big)\Big] - \Big[\log(2h) - \mathcal{H}\big(\text{Bin}(2h-1, \bar p)\big)\Big].$$

(Under MFM weights the first term becomes $-(1+\gamma)\log n + O(1)$; see Section 2.2.) Inserting the dense-regime entropy gives (4) with an explicit constant:

$$\mathbb{E}D \;\approx\; -\frac{K_0 - 1}{2}\log\frac{n}{K_0} - \log\frac{2n}{K_0\alpha} + C, \qquad C = \frac{K_0 - 2}{2}\log\big(2\pi e\, q(1-q)\big) + \tfrac{1}{2}\log\big(\pi e\, \bar p(1-\bar p)\big).$$

- For $p = 0.5$ and $q = 0.05$, $C = 0.27 - 0.105(K_0 - 2)$, which is small. For $n = 800$ and $K_0 = 4$ this gives $-13.9$, the value in Table 2.
- At fixed $h = 100$, each extra community adds one between-community term, $-\frac{1}{2}\log(h/(2\pi e\, q(1-q))) = -2.41$. This is the asymptotic value; the exact predictions of Table 2 give 2.43 per community.
- At fixed $K_0$, $\partial \mathbb{E}D/\partial \log n = -(K_0 - 1)/2 - 1 = -(K_0 + 1)/2$, which is $-2.5$ for $K_0 = 4$ (the exact predictions give $-2.45$ between $n = 200$ and 1600).

*Variance correction.* The true variance of $d_{iM}$ is $h[p(1-p) + q(1-q)]$, a fraction $r = [p(1-p) + q(1-q)]/[2\bar p(1-\bar p)]$ of the binomial one ($r = 0.75$ here). The noise in the $M$ term has mean $r/2$ rather than $\frac{1}{2}$, which lowers $\mathbb{E}D$ by about 0.13 nats. This is consistent with the measured means, which are mostly slightly below the predictions in Table 2.

## A.3 The average and the maximum over nodes

**Fluctuations.** Write $Y_{il} = n_l \mathrm{KL}(\hat\theta_{il} \| \bar\theta_{Ml})$ for the KL terms in (3). Under an invisible split, $Y_{il} \approx \frac{1}{2} Z_{il}^2$, with the $Z_{il}$ approximately independent standard normal variables, for the $K_0 - 2$ terms with $l \notin \{H, R\}$, whose counts are independent binomials. The $M$ term has the smaller variance noted above and is approximately $\frac{r}{2} Z_{iM}^2$. So, approximately,

$$D_i \;=\; \mathbb{E}D + (X_i - k) + (Y_{iM} - \tfrac{r}{2}), \qquad X_i \sim \text{Gamma}(k, 1), \qquad k = \tfrac{K_0 - 2}{2}.$$

Since $r < 1$, the $M$ term has a finite exponential moment, $\mathbb{E}e^{Y_{iM}} = (1-r)^{-1/2}$, and contributes a constant factor $(1-r)^{-1/2} e^{-r/2}$ to the average below, $e^{0.31}$ for $p = 0.5$ and $q = 0.05$. The heavy tail comes from the $K_0 - 2$ other terms, so there is none for $K_0 = 2$.

**Why the average, not the mean or the maximum.** (S1) opens a block at rate $\frac{1}{n}\sum_{i\in H\cup R} e^{D_i}/(1+e^{D_i})$ per update. The relevant quantity is therefore the *average opening odds*

$$\lambda \;=\; \frac{1}{h}\sum_{i\in H} e^{D_i}.$$

- *The graph average is misleading.* Since $\mathbb{E}\mathrm{BF} = 1$ (A.1), the expectation of $e^{D_i}$ over graphs is about $w_{\mathrm{new}}/w_M \approx K_0\alpha/(2n)$. It is dominated by rare graphs on which $i$ has atypical degrees, and it would wrongly suggest a cost of only $n^{-1}$.
- *Truncation.* For $K_0 \geq 3$, on a typical graph the variables $e^{X_i}$ are heavy-tailed, with $\mathbb{P}(e^X > t) \approx (\log t)^{k-1}/(\Gamma(k)\,t)$. Their sum over the $h$ nodes is governed by truncation at the typical maximum $x_{\max}$, which solves $h\mathbb{P}(X > x_{\max}) \approx 1$. The integrand $e^x$ times the Gamma$(k,1)$ density is $x^{k-1}/\Gamma(k)$, whose integral grows with the upper limit for every $k > 0$, so

$$\frac{1}{h}\sum_{i\in H} e^{X_i} \;\approx\; \int_0^{x_{\max}} \frac{x^{k-1}}{\Gamma(k)}\,dx \;=\; \frac{x_{\max}^k}{\Gamma(k+1)}.$$

- *Result.* $\log\lambda - \mathbb{E}D \approx 0.31 + k\log(x_{\max}/k) - \frac{1}{2}\log(2\pi k)$, with $x_{\max} \approx \log h + (k-1)\log\log h$ when $k \ll \log h$, and $x_{\max} \approx k + \sqrt{2k\log h}$ when $k \gg \log h$. In both cases

$$\log\lambda \;=\; \mathbb{E}D + O(K_0\log\log n) + O(\sqrt{K_0\log n}),$$

  which is the form of Lemma 2.3. The measured differences, about 0.28 for $K_0 = 2$ and 0.8 for $K_0 = 3$ (Table 10), are of this size.

**The maximum.** The largest $D_i$ exceeds $\mathbb{E}D$ by about $x_{\max} - k$. For fixed $K_0$ and large $h$ this is about $\log h$, not $O(K_0\log\log n)$; for large $K_0$ it is about $\sqrt{2k\log h}$. In Table 2 the maxima are taken over $h/2$ sampled nodes (all $h$ in the first block of rows) and exceed the means by 2.1 to 8.1 nats, in line with these sizes. For example, $K_0 = 4$ and $n = 1600$ give 5.6, against $\log 200 = 5.3$, and $K_0 = 16$ and $n = 1600$ give 8.1, against $\sqrt{2\cdot 7\cdot\log 50} = 7.4$ from the $K_0 - 2$ heavy-tailed terms alone. Two consequences:

- A bound on $\max_i D_i$ combined with a union bound over nodes loses a factor $h$. A proof must control the sum $\sum_i e^{D_i} = h\lambda$ directly.
- The few best-placed nodes open blocks far more often than the typical node. Such blocks still dissolve unless a nucleus forms (A.4).

## A.4 The nucleus

**Setup.** For a nucleus $\mathcal{N} \subset H$ with $|\mathcal{N}| = m$, let $C_{\mathcal{N}}$ be $C_{HR}$ with $\mathcal{N}$ moved into a new block. Let $S_m$ be the set of all such partitions, together with those built from $R$ in the same way, and let $F(m) = -\log\{\Pi_n(S_m)/\Pi_n(C_{HR})\}$ be its free energy. By A.3, $F(1) = -\log(2h\lambda)$.

**Prior and entropy.** Under DP$(\alpha)$, $p(C_{\mathcal{N}})/p(C_{HR}) = \alpha(m-1)!\,(n_M - m - 1)!/(n_M - 1)!$. Multiplying by the $2\binom{h}{m}$ choices of $\mathcal{N}$ and assuming $m^2 \ll h$,

$$\log\frac{p(S_m)}{p(C_{HR})} \;=\; \log\frac{2\alpha}{m} - m\log\frac{n_M}{h} + O\Big(\frac{m^2}{h}\Big).$$

Net of entropy, the prior's preferential attachment, which favours joining the large block $M$, costs $\log(n_M/h)$ per member.

**Likelihood.** Moving $\mathcal{N}$ creates two kinds of new pairs.

- $K_0 - 1$ pairs $(\mathcal{N}, l)$, each with $mn_l$ trials and no signal (Definition 2.1).
- The pair $(\mathcal{N},\mathcal{N})$, with $\binom{m}{2}$ trials at rate $p$. In $C_{HR}$ these trials were explained by $\bar{p}$, so the pair gains $\binom{m}{2}\kappa$ on average and pays a parameter cost of about $\frac{1}{2}\log\binom{m}{2}$.

The other parameters change little. The rate between $\mathcal{N}$ and $M\setminus\mathcal{N}$ differs from $\bar{p}$ by $m(p-q)/(2(2h-m))$, which contributes $O(m^3/h)$.

**Two competing effects of order** $\log m$**.**

- *Fixed nucleus.* For a fixed $\mathcal{N}$, each pair $(\mathcal{N}, l)$ costs $\frac{1}{2}\log(mn_l) + O(1)$. The cost grows by $\frac{1}{2}\log m$ per pair, or $\frac{K_0-1}{2}\log m$ in total.
- *Selection.* The sum over the $\binom{h}{m}$ nuclei favours nuclei whose members have aligned fluctuations. Let $z_{jl}$ be the standardised count from member $j$ to block $l$, so that the KL term of the pair $(\mathcal{N}, l)$ is $(\sum_{j\in\mathcal{N}} z_{jl})^2/(2m)$. The Gaussian identity $e^{v^2/(2m)} = (m/2\pi)^{1/2}\int e^{tv-mt^2/2}\,dt$ turns the sum over nuclei into an integral over a tilt $t \in \mathbb{R}^{K_0-1}$ of $e_m(e^{t\cdot z_1}, \dots, e^{t\cdot z_h})$, the elementary symmetric polynomial of degree $m$. For $|t|^2 < 2\log(h/m)$ this is about $(h^m/m!)\, e^{m|t|^2/2}$, so the integrand is flat in $t$, and the integral is

$$\frac{h^m}{m!}\; m^{(K_0-1)/2}\; \frac{(\log(h/m))^{(K_0-1)/2}}{\Gamma((K_0+1)/2)}.$$

  The factor $m^{(K_0-1)/2}$ cancels the growth of the $K_0 - 1$ parameter costs. At $m = 1$ the same computation reproduces the polylogarithmic factor of A.3.

**Result.** For $1 \le m \ll h^{1/3}$,

$$F(m) - F(1) \;=\; (m-1)\log\frac{n_M}{h} \;-\; \frac{m(m-1)}{2}\,\kappa \;+\; O(K_0 \log m).$$

With the selection effect the $\log m$ terms reduce to about $\log m + \frac{1}{2}\log\binom{m}{2}$. Without it they would be about $\frac{K_0+3}{2}\log m$. Section 2.3 uses only the bound $O(K_0 \log m)$, which covers both.

- *Location of the maximum.* $\log(n_M/h)/\kappa \le m^* \le \log(n_M/h)/\kappa + O\big(1 + \sqrt{K_0/\kappa}\big)$.
- *Height.* $F(m^*) - F(1) \le (\log(n_M/h))^2/(2\kappa) + O(\log(n_M/h)) + O(K_0 \log m^*)$.
- *Independence of $n$.* Both quantities are independent of $n$ in the dense regime.

**Two cases.**

- *Merged pair, $n_M/h = 2$.* For $p = 0.5$ and $q = 0.05$, $\bar{p} = 0.275$ and $\kappa = 0.113$. The leading terms give $m^* \approx \log 2/\kappa \approx 6$ and a height of about $(\log 2)^2/(2\kappa) \approx 2.1$ nats. The greedy paths of Appendix B reach their lowest point at $m = 4$ to 8 in the dense experiments (Table 9). The agreement with $m^* \approx 6$ may be coincidental, since a single path ignores entropy (Appendix B.3).
- *Dilution.* A block holding $r$ communities has $n_M/h = r$ and $\bar{p} \approx (p + (r-1)q)/r$. For $r = 16$ and the same $p, q$: $\bar{p} = 0.078$, $\kappa = 0.62$, and $(\log 16)^2/(2\kappa) \approx 6.2$ nats.

## A.5 From free energies to times

**Birth–death reduction.** Project the chain near $C_{HR}$ onto the nucleus size $m$, with weights $\pi(m) = \Pi_n(S_m)$ and $\pi(0) \approx \Pi_n(C_{HR})$. Detailed balance gives $\pi(m)u(m) = \pi(m+1)d(m+1)$, where

- $u(m)$ is the probability per update that the nucleus grows;
- $d(m) \approx m/n$ is the probability per update that one of its $m$ members is selected and leaves. Below $m^*$ a selected member leaves with probability close to one.

The passage-time formula for birth–death chains, together with $\pi(0) \gg \pi(m)$ for $m \ge 1$, gives

$$\mathbb{E}\tau_{\text{sep}} \;\approx\; \sum_{m\ge 1} \frac{\pi(0)}{\pi(m)\,d(m)} \;\approx\; \frac{n}{m^*}\, e^{F(m^*)}\sqrt{2\pi/\kappa}.$$

The last factor is the width of the peak of $F$; it is a crude approximation here, since $\sqrt{2\pi/\kappa} \approx 7.5$ exceeds $m^* \approx 6$. Using $F(1) = -\log(2h\lambda)$,

$$\mathbb{E}\tau_{\text{sep}} \;\approx\; \frac{K_0}{2\lambda\rho}, \qquad \rho^{-1} = \frac{e^{F(m^*)-F(1)}\sqrt{2\pi/\kappa}}{m^*}.$$

Here $\rho$ is, roughly, the probability that a newly opened block grows to $m^*$ before it dissolves. In this heuristic it does not depend on $n$. Combined with $\lambda = e^{\mathbb{E}D}(\log n)^{O(K_0)}$ (A.3) and (4), this gives the exponent law (5).

**What the heuristic does and does not predict.** It predicts the exponent of $n$. It does not predict constants. The $O(K_0 \log m^*)$ terms in $F$, the width factor and the departure probabilities can each change $\rho$ by a factor

of ten or more at the sizes simulated here. The simulations add a further caution: the measured $\tau\lambda$ still decreases with $n$ (Table 3), which is consistent with $\rho$ growing with $n$, so the apparent exponent at accessible sizes is below the asymptotic one. The terms of order $m^{*2}/h$ and $m^{*3}/h$ dropped above are one candidate cause: with $m^* \approx 6$, $m^{*3}/h$ is about 4 at $h = 50$ and 0.3 at $h = 800$. They act in opposite directions (the first penalises large nuclei, the second rewards them), and which dominates has not been worked out. Heuristically the lower-bound side is robust: separation requires at least one opening, which takes about $K_0/(2\lambda)$ updates, so $\mathbb{E}\tau_{\text{sep}} \gtrsim K_0/(2\lambda)$ whatever $\rho$ does. A proof of this still needs the summation over $\mathcal{B}$ below.

**The lower bound.** A standard escape bound holds for a reversible chain started at $x \in \mathcal{B}$:

$$\mathbb{P}_x(\tau_{\mathcal{B}^c} \le t) \;\le\; t\,\frac{Q(\mathcal{B},\mathcal{B}^c)}{\Pi_n(x)}, \qquad Q(\mathcal{B},\mathcal{B}^c) = \sum_{y\in\mathcal{B},\, y'\notin\mathcal{B}} \Pi_n(y)P(y,y').$$

*Proof sketch.* Start the chain from $\Pi_n$ restricted to $\mathcal{B}$, and kill it when it leaves $\mathcal{B}$. At every step its law is dominated by $\Pi_n$ restricted to $\mathcal{B}$, so it exits at that step with probability at most $Q(\mathcal{B},\mathcal{B}^c)/\Pi_n(\mathcal{B})$. Starting at $x$ instead costs at most a factor $\Pi_n(\mathcal{B})/\Pi_n(x)$.

*Choice of $\mathcal{B}$.* The set must contain $C_{HR}$ and its basin, and every partition that counts as separated must lie outside it, so that $\tau_{\text{sep}} \ge \tau_{\mathcal{B}^c}$. Lemma 2.4 takes

$$\mathcal{B} = \Big\{C :\ \text{at most one block of } C \text{ contains } m^* \text{ or more nodes of } H\cup R\Big\}.$$

This set has the required properties:

- $C_{HR} \in \mathcal{B}$;
- every partition counted by $\tau_{\text{sep}}$ lies outside, since it has two blocks of size at least $n/(2K_0)$ with majorities $H$ and $R$, each holding at least $n/(2K_0^2) \ge m^*$ nodes of $H \cup R$;
- every exit creates a second block, pure or mixed, with $m^*$ nodes of $H \cup R$, that is, a nucleus of size $m^*$. A mixed nucleus has less internal signal than a pure one, so pure nuclei dominate the flux.

Heuristically, $Q(\mathcal{B},\mathcal{B}^c) \approx \pi(m^*)\,d(m^*)$, so

$$\frac{Q(\mathcal{B},\mathcal{B}^c)}{\Pi_n(C_{HR})} \;\approx\; \frac{m^*}{n}\,e^{-F(m^*)} \;=\; \frac{2m^*}{K_0}\,\lambda\,e^{-(F(m^*)-F(1))} \;\le\; n^{-\frac{K_0+1}{2}}(\log n)^{O(K_0)}e^{O(K_0\log K_0)}.$$

For fixed $K_0$ this is $n^{-(K_0+1)/2+o(1)}$, and with $t = n^{(K_0+1)/2-\epsilon}$ the escape bound gives the lower half of Target 2.1. Two tempting alternatives fail.

- "No block contains at least $m^*$ nodes of $H$ and at most $m^*/2$ nodes of other communities" contains partitions that agree with $C_\star$ except for $m^*/2+1$ nodes of a third community placed in $H$'s block. Those carry far more posterior mass than $C_{HR}$, and the bound becomes useless.
- "Every block has $\big||c\cap H| - |c\cap R|\big| < m^*$" contains partitions that split $M$ into two nearly balanced halves. Because $|H| = |R|$, the halves have opposite majorities unless they tie, so these partitions count as separated under Definition 2.2, and $\tau_{\text{sep}} \ge \tau_{\mathcal{B}^c}$ fails.

## A.6 The underestimation law

Start from the one-block partition. The simulations show the chain peeling off one community at a time: the final block sizes of the companion runs were $[599, 200, 1]$, $[400, 200, 200]$, $[800]$, $[700, 99, 1]$, $[799, 1]$ and $[750, 50]$. So suppose that at some stage the chain holds $j$ blocks: $j-1$ single communities and one block $M_j$ holding the remaining $r = K_0 - j + 1$ communities.

- *No signal.* For a node of $M_j$, its rate into $M_j$ equals $M_j$'s internal density, by the computation of A.2 with equal community sizes, and its rate into every other block equals the corresponding block rate.
- *Cost.* Opening therefore creates $j$ new parameters without signal, $j-1$ of them from $h$ trials and one from $rh$ trials, and the DP weight ratio is about $\alpha/(rh)$:

$$\mathbb{E}D \approx -\frac{j}{2}\log h - \log\frac{rh}{\alpha} + O(j + \log r).$$

- *Time.* The next successful opening takes about $n^{(j+2)/2}(\log n)^{O(j)}e^{O(j\log j)}/\rho_j$ updates for fixed $K_0$. Here $\rho_j$ contains the dilution term $(\log r)^2/(2\kappa_r)$ of A.4, with $r = K_0 - j + 1$, which can only slow things down.

Reaching $k$ blocks requires the stage $j = k-1$, which takes about $n^{(k+1)/2}$ updates. With a budget of $T = n^\beta$ updates, $(k+1)/2 \le \beta + o(1)$, so

$$k \;\le\; 2\beta - 1 + o(1) \;=\; 2(\beta-1) + O(1),$$

whatever $K_0$ is (with $h = n/K_0$, the exponent of $n$ is unchanged as long as $K_0$ is fixed). For the long runs of Table 4, $T = 8\times 10^5 = n^{2.03}$, which gives $k \le 3$. Two communities were found. Under MFM weights each stage costs an extra $n^\gamma$, which gives $k \le 2(\beta-\gamma) - 1 + o(1)$.

### A.7 Extra blocks at the truth

Repeat A.4 at $C_\star$, with a new block $\mathcal{N}$ made of $m$ nodes of one community $c$.

- *Prior and entropy.* Exactly, $\binom{h}{m}\,\alpha\,(m-1)!\,(h-m-1)!/(h-1)! = \alpha h/(m(h-m))$. This is about $\alpha/m$ for small $m$, so the DP prior does not penalise an extra small block.
- *Likelihood.* The likelihood does. There are $K_0$ new pairs $(\mathcal{N}, l)$ without signal, and the pair $(\mathcal{N},\mathcal{N})$ also has none, because $c$'s internal rate is $p$. With the selection effect of A.4, the sum over nuclei of size $m$ costs about $h^{-K_0/2}(\log h)^{K_0/2}/\Gamma(K_0/2+1)$, up to $e^{O(K_0)}$.
- *Total.* The posterior odds of one extra block, summed over sizes and over the $K_0$ communities, are $O\big(K_0\alpha h^{-K_0/2}(\log n)^{O(K_0)}\big)$. More extra blocks cost more. This is the reasoning behind Target 2.7.
- *Numbers.* For one-node extra blocks at $n = 800$, the odds summed over all nodes are $e^{-3.2}$, $e^{-5.8}$, $e^{-13.0}$ and $e^{-24.7}$ for $K_0 = 2, 4, 8, 16$. The leading term $\log(K_0\alpha h^{-K_0/2})$ gives $-5.3$, $-9.2$, $-16.3$ and $-28.5$. The offset grows slowly, from 2.1 to 3.8 nats, as expected from the truncated sums of A.3. The sums are dominated by a few nodes: estimates from every fourth node gave $-7.0$, $-13.9$ and $-26.0$ for $K_0 = 4, 8, 16$.

*Contrast with mixtures.* For a conjugate Gaussian location mixture in fixed dimension, a one-point cluster costs only $O(1)$ in likelihood. Summed over points, the extra-cluster odds therefore stay of order $\alpha$, the mechanism behind the inconsistency of Dirichlet-process mixtures for the number of clusters (S8). In the SBM, a one-node block still has $n/K_0$ trials for each new parameter, so the summed odds are of order $h^{-K_0/2}$.

### A.8 Visible splits

Let $\delta_l = |B_{\star,Hl} - B_{\star,Rl}|$ and $\theta_l = B_{\star,Hl}$. For $i \in H$ and $l \notin \{H, R\}$, the node's rate is $\theta_l$ while $M$'s rate is $\bar\theta_{Ml} = (B_{\star,Hl} + B_{\star,Rl})/2$. The KL term then has mean

$$\mathbb{E}Y_{il} \;\approx\; \tfrac12 + n_l \mathrm{KL}(\theta_l \| \bar\theta_{Ml}) \;\approx\; \tfrac12 + \frac{n_l\,\delta_l^2}{8\,\theta_l(1-\theta_l)}.$$

If $B_{\star,HH} \ne B_{\star,RR}$, the $M$ term contributes similarly.

- *Opening.* A typical node opens favourably once $\sum_l n_l\delta_l^2/(8\theta_l(1-\theta_l))$ exceeds about $\frac{K_0+1}{2}\log n$. This gives the crossover of Section 2.3 and part (b) of Target 2.4.
- *Growth.* Each additional member of a nucleus adds the signal $S = \sum_l n_l \mathrm{KL}(\theta_l\|\bar\theta_{Ml})$, against the prior's $\log(n_M/h)$. Once $S > \log(n_M/h) + O(K_0)$, the nucleus has a positive drift from $m = 1$. This condition is much weaker than the opening condition.
- *Experiment E9.* None of the 50 sampled nodes had positive opening odds (maximum $-1.8$), yet (S1) found all eight communities in the first sweep, with one extra small block. A rare opening is followed by downhill growth.
- *Intermediate regime.* For $1 \ll S \ll K_0 \log n$, the mean KL terms lower the opening cost, and the exponent should fall continuously from $(K_0+1)/2$ as $S/\log n$ grows. This is part (c) of Target 2.4.

### A.9 Spectral splits

Take a block that is the union of $r$ communities of size $h$, with rates $p$ within and $q$ between communities.

- *Structure.* After centring by the block's density, the expected adjacency matrix is $(p-q)\big(\sum_c \mathbf{1}_c\mathbf{1}_c^\top - \frac{1}{r}\mathbf{1}\mathbf{1}^\top\big)$, up to the diagonal. Its nonzero eigenvalue is $(p-q)h$, with multiplicity $r-1$ and eigenspace $\{\sum_c a_c\mathbf{1}_c : \sum_c a_c = 0\}$.
- *Noise.* In the dense regime the noise matrix has operator norm $O(\sqrt{rh})$ with high probability.
- *The case $r = 2$.* The eigenspace is spanned by $\mathbf{1}_H - \mathbf{1}_R$. Davis–Kahan puts the leading eigenvector within $O(h^{-1/2})$ of it in $\ell_2$, and entrywise bounds (G5) give exact signs with high probability. So the split proposal of (S2) recovers $H$ and $R$.
- *The case $r \geq 3$.* The eigenvalue is degenerate, and the noise selects a vector $\sum_c a_c\mathbf{1}_c$ in the eigenspace. The sign split keeps every community whose $|a_c|$ is not small on one side, and may cut a community whose $a_c$ is close to zero, as in Lemma 2.6.
- *Acceptance.* Splitting a merged pair gains the full log-posterior gap, between about 520 and 45,000 nats in the experiments (Table 9). The proposal ratio is only polynomial in $K$, so the split is accepted. A split of a true community has no signal and is rejected.

# B Simulation details

The scripts and raw logs accompany this document as `p3c_simulations.zip`. The runs labelled "companion" were made for the P3 document; E1–E10 were made for this one.

## B.1 Model, priors and implementation

- *Graphs.* Communities have equal size $h = n/K_0$, and nodes are ordered by community. Each $A_{ij}$, $i < j$, is drawn from uniform variates generated with the graph seed. E9 and E10 use a heterogeneous $B_\star$ (Table 8).
- *Priors.* Beta$(1,1)$ for every block-pair probability and DP$(\alpha = 1)$ for the partition.
- *Log posterior*, up to a constant: $K \log \alpha + \sum_k \log \Gamma(n_k) + \sum_{k \le l} \log \mathrm{B}(1 + E_{kl}, 1 + N_{kl} - E_{kl})$.
- *State.* Labels, block sizes and the matrix of block edge counts are updated incrementally. An emptied block is deleted by moving the last label into its place. The incremental bookkeeping was checked against a full recomputation of the log posterior.

## B.2 Samplers

- (S1). As in Section 2.2, with DP weights. Each sweep visits all $n$ nodes once, in a fresh uniformly random order, so one sweep is $n$ updates. The chain's random seed is the graph seed plus 777 (E1–E10), 1000 (companion runs from the merged pair) or 2000 (companion runs from one block).
- (S2). Three split–merge proposals precede every sweep of (S1).
  - *Split.* The leading eigenvector, for the eigenvalue of largest modulus, is computed for the block's adjacency submatrix minus its mean off-diagonal density, with zero diagonal. Blocks with fewer than four nodes are not split, and a split that leaves one side empty is not proposed.
  - *Merge.* Two distinct blocks are chosen uniformly at random. The proposal is rejected unless the spectral split of their union returns exactly that pair.
  - *Acceptance.* The Metropolis–Hastings ratio uses the proposal probabilities of Section 2.2. Because the split is a deterministic function of the block, and the merge check guarantees that the reverse split is available, each proposal satisfies detailed balance, with one exception. When $K = 1$ the code always proposes a split, but the ratio still uses the forward probability $\frac{1}{2}/\#\{\text{eligible blocks}\}$; merges into $K = 1$ are treated likewise. Detailed balance between $K = 1$ and $K = 2$ is therefore off by a factor of 2, which is immaterial here because the log-posterior gaps exceed 500 nats.
  - *Composition.* Proposals and sweeps are applied in a fixed order, so the combined kernel leaves the posterior invariant but is not reversible (compare Target 2.5).

## B.3 Diagnostics

- *Opening log-odds $D_i$.* These are exact: the difference between the log-weights of "new block" and "join $M$" in (S1), computed at $C_{HR}$ after removing $i$. They are computed for every second node of community 1 in E1–E10 ($h/2$ nodes), and for all $h$ nodes in the companion runs. The "mean" and "max" in Table 2 are over these nodes.
- *Extra-singleton odds at the truth.* The same quantity at $C_\star$. Table 9 gives the mean over every eighth node (every fourth in the companion rows). The summed odds in Target 2.7 and Appendix A.7 use all $n$ nodes.
- *Greedy barrier.* Start from $C_{HR}$ and move the nodes of community 1 into a new block one at a time, each time choosing the node that maximises the log posterior. Let $G(m)$ be the log posterior after $m$ moves minus that of $C_{HR}$. The barrier is $-\min_m G(m)$, attained at the reported $m$.
  - This is the depth of one explicit path. It bounds from above the barrier of the best path of this kind.
  - It is not the free-energy barrier of Appendix A.4, because a single path ignores entropy. It can exceed the opening cost by a wide margin.
- *Gap.* $\log \Pi_n(C_\star) - \log \Pi_n(C_{HR})$.

## B.4 Success criteria

- *Communities found.* The number of distinct true communities that are the majority of some block of size at least $h/2$.
- *Hitting time.* The first sweep after which all $K_0$ communities are found. It is checked after every sweep.
- *Companion runs from the merged pair.* These used the first sweep with at least $K_0$ blocks of size at least $h/2$, without the majority check. The final block sizes recorded there show whole communities.
- *Purity check.* Runs were repeated with the same seeds, and the blocks inspected at the hitting time and twenty sweeps later.
  - Twelve $K_0 = 2$ runs, including all four hits within six sweeps at $n = 1600$. In 11 of them each community had a *pure* block at the hitting time, although in two of these a large mixed third block was also present (183 nodes with a 58% majority, and 534 nodes with 52%). In the remaining run ($n = 200$) one block was pure and the other was mixed, with a 63% majority. Twenty sweeps later all twelve were at the truth, up to at most eight nodes.
  - Six $K_0 = 3$ runs, including all three hits within six sweeps. At the hitting time every community was the majority of a block with purity at least 68%. Twenty sweeps later five were exactly at the truth, and one had 13 nodes outside the three large blocks.

  So fast hits are genuine separations.

## B.5 Experiments

| ID | Purpose | $n$ | $K_0$ | Start | Sampler, budget | Seeds |
|---|---|---|---|---|---|---|
| comp. | cost, barrier, separation | 800 | 2, 4, 8, 16 | merged; truth | (S1), 30 sweeps | 1, 2 |
| comp. | underestimation | 800 | 4, 8, 16 | one block | (S1), 100 | 1, 2 |
| E1 | scaling in $K_0$ at $h = 100$ | $100K_0$ | 2, 4, 8, 16 | merged; truth | (S1), 50; 20 | 1–3; 1 |
| E2 | scaling in $n$ at $K_0 = 4$ | 200–1600 | 4 | merged | (S1), 100 | 1–4 |
| E3 | hitting times | 100–800 | 2 | one block | (S1), $\leq 300$ | 1–8 |
| E3b | more hitting times | 100–1600 | 2 | one block | (S1), $\leq 1000$ | 9–40; 1–24 at $n = 1600$ |
| E4 | logarithmic degree | 800 | 2, 4, 8 | merged | (S1), 50 | 1, 2 |
| E5 | split–merge | 800 | 4, 8, 16 | merged; one block | (S2), 30 | 1, 2 |
| E6 | long runs | 800 | 16 | one block | (S1), 1000 | 1, 2 |
| E7 | hitting times | 150–1200 | 3 | merged | (S1), $\leq 4000$ | 1–16; 1–12 at $n = 1200$ |
| E9 | visible split | 800 | 8 | merged | (S1), 30 | graph 1; chains 1, 2 |
| E10 | invisible split | 800 | 8 | merged | (S1) and (S2), 30 | graph 1; chains 1, 2 |

**Table 8:** The experiments. Unless stated otherwise, $p = 0.5$ and $q = 0.05$, and seed $s$ means graph seed $s$ with the chain seed of Appendix B.2. There is no E8. *E4:* $p = \tilde{a}\log n/n$ and $q = \log n/n$ with $\tilde{a} = (1 + 1.5\sqrt{K_0})^2$, so that $(\sqrt{\tilde{a}} - 1)^2 = 2.25K_0$; this gives $p = 0.081, 0.134, 0.230$ for $K_0 = 2, 4, 8$, and $q = 0.0084$. *E9:* off-diagonal entries of $B_\star$ are independent $U(0.05, 0.3)$ and diagonal entries $U(0.3, 0.6)$, drawn with seed 5; for the merged pair, $B_{\star,00} = 0.33$, $B_{\star,11} = 0.51$ and $B_{\star,01} = 0.25$. *E10:* as E9, but row and column 1 are set equal to row and column 0 outside $\{0, 1\}$, with $B_{\star,00} = B_{\star,11} = 0.5$ and $B_{\star,01} = 0.1$, so that communities 0 and 1 form an invisible split.

**Other results** (Section 2.4).

- *Separation from the merged pair by* (S1) (Table 2). $K_0 = 2$: 1 of 2 companion runs (sweep 4), 1 of 3 in E1 (sweep 11), 1 of 2 in E4 (sweep 26). $K_0 \geq 4$: none.
- *Runs from the truth* (E1 and companion). (S1) stayed at $C_\star$ in all of them.
- *Split–merge* (E5, E10). See Tables 5 and 6.
- *Underestimation* (companion and E6). See Table 4.

## B.6 Hitting times

- *Opening rate.* $-\log\lambda$ has slopes 1.52 and 1.94 on $\log n$, against 1.50 and 1.98 for $-\mathbb{E}D$. The difference $\log\lambda - \overline{D_i}$ is about 0.28 for $K_0 = 2$ and 0.8 for $K_0 = 3$, and nearly constant in $n$. It is the gain from

| Experiment | $n$ | $K_0$ | opening mean (max) | greedy barrier (at $m$) | gap | extra singleton at truth |
|---|---|---|---|---|---|---|
| companion | 800 | 2 | −9.5 (−5.5) | 12.5 (6) | 44,753 | −11.3 |
| | 800 | 4 | −14.2 (−9.4) | 17.3 (4) | 11,153 | −15.5 |
| | 800 | 8 | −22.0 (−15.5) | 23.1 (7) | 2,682 | −23.3 |
| | 800 | 16 | −36.1 (−29.9) | 45.4 (8) | 522 | −36.9 |
| E1 | 200 | 2 | −7.5 (−5.4) | 10.3 (4) | 2,687 | −8.7 |
| | 400 | 4 | −12.2 (−9.2) | 17.0 (6) | 2,644 | −13.3 |
| | 800 | 8 | −21.9 (−15.5) | 23.1 (7) | 2,682 | −23.1 |
| | 1600 | 16 | −41.9 (−33.8) | 52.8 (7) | 2,640 | −42.5 |
| E2 | 200 | 4 | −10.5 (−7.3) | 9.5 (4) | 639 | −11.7 |
| | 800 | 4 | −14.0 (−10.4) | 17.3 (4) | 11,153 | −15.3 |
| | 1600 | 4 | −15.8 (−10.2) | 23.3 (7) | 44,958 | −17.6 |
| E4 | 800 | 2 | −10.1 (−6.6) | 25.2 (19) | 5,234 | −12.7 |
| | 800 | 4 | −16.6 (−10.4) | 19.1 (11) | 2,539 | −18.7 |
| | 800 | 8 | −27.8 (−21.8) | 30.7 (5) | 1,301 | −30.2 |

**Table 9:** Static diagnostics (graph seed 1), in nats. E2 at $n = 400$ is the same graph as E1 at $K_0 = 4$. The companion graphs at $K_0 = 4$ and 8 are the same as E2 at $n = 800$ and E1 at $K_0 = 8$; the opening statistics differ slightly because the companion runs use all nodes of community 1. The last column is a mean over sampled nodes (every fourth node in the companion runs, every eighth in E1–E4).

fluctuations computed in A.3: about 0.31 from the $M$ term, which has reduced variance, plus a heavy-tail term that is present only for $K_0 \geq 3$.

- *Hitting times.* The mean is the quantity in (5). Its slopes are 0.39 and 0.56 in sweeps, that is 1.39 and 1.56 in updates. The geometric mean is pulled down by fast hits, which occur at every $n$. Hits within six sweeps are more frequent than an exponential law with the observed means would give (3 of 32 runs at $K_0 = 3$, $n = 300$ and 600, against about 0.2 expected), so the hitting law is not exponential. The graphs of these runs have $\log \lambda$ only 0.1 to 0.8 above the median.
- *Nucleus factor.* The slope of log of the mean of $\tau\lambda$ on $\log n$ is $-0.13$ for $K_0 = 2$ and $-0.45$ for $K_0 = 3$. This is consistent with $\rho$ growing with $n$ in this range, more for $K_0 = 3$, but $\rho$ itself was not measured.
- *Interpretation.* The levels $\tau\lambda \approx e^{1.5}$ to $e^{3.1}$ are well above the floor of about $K_0/2$ set by the first opening, so the trend can continue over a few more doublings of $n$; for $K_0 = 3$ it would reach the floor near $n \approx 6 \times 10^4$. The asymptotic exponent cannot be reached at the sizes simulated here. Candidate causes are listed in Section 2.4.

## B.7 Run counts and reproducibility

- *Determinism.* A run is determined by $(n, K_0, p, q)$, the graph seed and the chain seed. E1 at $K_0 = 4$ and E2 at $n = 400$ share these for seeds 1–3, so the E1 runs are the first 50 sweeps of the E2 runs.
- *Merged-pair runs of* (S1) *with* $K_0 \geq 4$. There are 6 companion runs, 9 in E1, 16 in E2, 4 in E4 and 2 in E10, a total of 37. Without the 3 duplicates, 34 distinct runs remain. None separated the pair.
- *Compute.* Everything ran on two CPU cores. The longest single run, $K_0 = 3$ at $n = 1200$, took under four minutes.
- *Scripts.*
  - model and (S1): `sbm_k.py`; (S2) and diagnostics: `sbm_sm.py`;
  - experiments: `sbm_exp2.py`, `sbm_e7.py`, `sbm_e10.py`, and the companion scripts `sbm_k_exp.py` and `sbm_k_one.py`;
  - analysis: `predict_open.py` (Table 2), `hit_stats.py` and `lambda_check.py` (Tables 3 and 10);
  - checks: `check_purity.py` and `check_purity3.py` (purity), `truth_singletons.py` (extra-singleton odds);
  - outputs: `logs/`, including `companion_results.txt` for the companion runs.

| $K_0$ | $n$ | runs | median | mean | geometric mean | range | mean $\log \lambda$ | mean $D_i$ |
|---|---|---|---|---|---|---|---|---|
| 2 | 100 | 40 | 25.5 | 34.1 | 26.5 | 6–118 | −6.12 | −6.40 |
| | 200 | 40 | 25.0 | 40.4 | 25.4 | 3–163 | −7.18 | −7.46 |
| | 400 | 40 | 33.0 | 47.0 | 31.9 | 6–184 | −8.23 | −8.50 |
| | 800 | 40 | 53.5 | 98.6 | 48.8 | 5–455 | −9.28 | −9.55 |
| | 1600 | 24 | 64.0 | 83.8 | 45.4 | 4–452 | −10.32 | −10.59 |
| 3 | 150 | 16 | 135.5 | 282.2 | 163.7 | 16–1159 | −7.76 | −8.54 |
| | 300 | 16 | 523.0 | 814.9 | 298.0 | 4–3486 | −9.09 | −9.89 |
| | 600 | 16 | 515.5 | 840.9 | 371.6 | 4–2623 | −10.40 | −11.26 |
| | 1200 | 12 | 759.5 | 1025.2 | 661.2 | 86–2723 | −11.81 | −12.64 |

| Slope on $\log n$, in sweeps (95% bootstrap interval) | mean | median | geometric mean |
|---|---|---|---|
| $K_0 = 2$ (prediction 0.5) | 0.39 (0.22–0.54) | 0.38 (0.06–0.55) | 0.25 (0.08–0.42) |
| $K_0 = 3$ (prediction 0.98) | 0.56 (0.22–0.93) | 0.74 (0.25–1.12) | 0.64 (0.25–1.00) |

**Table 10:** Hitting times in sweeps. No run was censored. $\lambda$ is the exact average of $e^{D_i}$ over the nodes of $H \cup R$ at $C_{HR}$ for the graph of each run, and "mean $D_i$" the average of $D_i$ itself. The predictions use the exact $\mathbb{E}D$. The bootstrap resamples runs within each $n$ (20,000 replicates).

## B.8 Limitations

- *Sizes.* $n \leq 1600$, and one graph per setting for the static diagnostics.
- *Seeds.* Two to four runs per setting, except the hitting-time experiments.
- *Scan.* Random-permutation scans rather than independent random updates. This changes constants only.
- *Priors and models.* DP(1) and Beta$(1, 1)$ only, no MFM, no degree correction. The planted partition, plus one heterogeneous example of each kind (E9, E10).
- *Exponents.* Hitting times were measured only for $K_0 = 2$ and 3, and their exponents are below the predictions at these sizes; the survival probability $\rho$ was not measured directly. For $K_0 \geq 4$ the evidence is that no run separated, which agrees with the predicted times but does not measure them.
- *MFM.* The predicted shift of all exponents by $\gamma$ under MFM weights was not tested.
- *Criterion.* "Found" needs blocks of size at least $h/2$ with distinct majorities. The purity check supports it, but it was not run on every hit.